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RESEARCH ARTICLE

Flow over periodic hills: an experimental study

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Abstract Two-dimensional flow over periodically arranged hills was investigated experimentally in a water channel. Two-dimensional particle image velocimetry (PIV) and one-dimensional laser Doppler anemometry (LDA) measurements were undertaken at four Reynolds numbers ($5{,}600 \leq Re \leq 37{,}000$). Two-dimensional PIV field measurements were thoroughly validated by means of point-by-point 1D LDA measurements at certain positions of the flow. A detailed study of the periodicity and the homogeneity was undertaken, which demonstrates that the flow can be regarded as two-dimensional and periodic for $Re \geq 10{,}000$. We found a decreasing reattachment length with increasing Reynolds number. This is connected to a higher momentum in the near-wall zone close to flow separation which comes from the velocity speed up above the obstacle. This leads to a velocity overshoot directly above the hill crest which increases with Reynolds number as the inner layer depth decreases. The flow speed up above that layer is independent of the Reynolds number which supports the assumption of inviscid flow disturbance in the outer layer usually made in asymptotic theory for flow over small hills.

1 Introduction

Separated flows are characterized by a fast thickening of the wall layer and large wall-normal velocity components. This induces a strong interaction between the near-wall and the external flow, which, in general, can lead to dramatic changes in the aerodynamic or transport properties of the flow. The point or line of separation can be fixed in space by sharp edges. Flow separation from smoothly contoured walls, however, can be very sensitive to external or internal flow conditions, such as properties of the oncoming flow, wall roughness, turbulence level of the outer flow, development of the streamwise pressure gradient, and many more. It might be stated that these complicated mechanisms are the reason that, up to now, no generally accepted theory for separation prediction has been found.

Flow separation from smoothly contoured obstacles, such as hills, occurs when the curvature of the surface or the height-to-length value, respectively, exceeds a certain value. In such cases, the turbulent boundary layer above such structures can be largely modified, which makes modeling extremely difficult (Athanassiadou and Castro 2001). The interest in separated flows over and past smoothly contoured obstacles comes from several disciplines including meteorology, aerodynamics, and process technology. The flow considered here can be seen as a prototype including a richness of various phenomena connected with flows over smoothly contoured 2D obstacles. It has common features with previous studies on flows over hill-like geometries such as wavy walls or isolated hills. It can serve in complementing the findings of these previous studies and in clarifying the influence of shape, aspect ratio, boundary layer characteristics, and others on the behavior of the separation zone. In the following, previous studies of related flow cases are briefly sketched before the geometry and the boundary conditions with their impact on two-dimensionality and periodicity of the flow are discussed.

A literature research with the keywords 'flow over wavy walls' takes the interested reader back to the year 1937,

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when Motzfeld (1937) did investigations in a wind tunnel. He measured velocities, pressure distributions, and the drag behavior of wind over modeled water waves.[1] Many works have followed, but as stated in studies like 'large-scale structures in a developed flow over a wavy wall' by Günther and v. Rohr (2003), the complexity of these flows, namely separation from a smoothly curved surface, recirculation, and reattachment, leaves questions unanswered to date.

Considerable theoretical understanding of the flow over small hills was obtained by linear analysis and asymptotic analysis (e.g. Jackson and Hunt 1975; Sykes 1980; Hunt et al. 1988; Belcher et al. 1993) that predicts mean velocity changes by small hills. With this analysis, it was possible to predict speed-up ratios over the crest of a single hill (Gong and Ibbetson 1989) and periodically arranged hills Athanassiadou and Castro (2001), both with rough surfaces. These studies confirm that rapid distortion theory was able to predict the attenuation of the streamwise component of turbulence intensity, especially in the outer layer.

Investigations into small- and large-amplitude sinusoidal waves with smooth surfaces were conducted by Zilker et al. (1977) and Zilker and Hanratty (1979) in a water channel. Ten wave trains were intended to assure the periodicity of the flow in the streamwise direction from the eighth period on.[2] The wall shear stresses were recorded by means of electro-chemical probes, and velocities were measured with thermal coated films. These devices were not sophisticated enough to yield more than a qualitative insight in this complex flow. Buckles et al. (1984) used the same channel but high-resolution pressure cells as well as laser Doppler anemometry (LDA), to observe the features of the flow. The periodicity of the flow was shown by velocity and rms profiles above the eighth and ninth wave crests. Hence, instantaneous information for this complex turbulent flow for a wave height–based Reynolds number of $Re = 12{,}000$ was available. Later et al. (1996) investigated Reynolds stress distributions and turbulence production in this flow. The group around von Rohr concentrated in a number of publications on coherent structures in the flow over a wavy wall and the influence of wavy geometry on them (e.g. Kruse et al. 2006; Kuhn et al. 2007; Günther and v. Rohr 2003). The influence of roughness on the turbulence structure above sinusoidal waves was investigated by Gong et al. (1996) who also provided a comparison to a large eddy simulation of this flow.

In 1993, an article was published by Almeida et al. (1993), who installed two-dimensional hills in their water channel that correspond to the symmetry axis of a three-dimensional hill introduced by Hunt and Snyder (1980) and investigated separation, recirculation, and reattachment. The smoothly shaped hills of height h were $3.857h$ long and well defined by six polynomials of third degree. They constricted the $6.07h$ channel by about one sixth. Two cases were studied: The flow over a single hill and an array of ten hills. Almeida et al. chose an inter-hill distance of $4.5h$ and a lateral extent of the domain of $4.5h$ as well. The measurements with an LDA system were carried out at $Re = 1.0 \times 10^4$ (based on hill height, see Eq. 1) between the hills seven and eight. Doubts arose concerning the true periodicity of the flow, and the side-wall effect was unknown. However, the experiment by Almeida et al. became basis for a testcase of the *ERCOFTAC/IAHR*-Workshop in 1995 (Rodi et al. 1995).

The flow considered here follows Mellen et al. (2000) who took advantage of and adjusted Almeida et al.'s geometry to fit numerical needs. Without touching the shape of the hill, the channel height was down-sized by a factor of two to save computational costs and the distance between succeeding hills was doubled to allow for natural reattachment on the flat bottom plate (see Fig. 1). Note that in Almeida et al.'s single-hill experiment, the flow reattaches at 4.82 hill heights after the hill crest which lies well before the consecutive hill starts. Furthermore, periodicity was applied in the streamwise and in the spanwise direction to keep the numerical cost affordable and to obtain well-defined flow conditions. The Reynolds number, which is based on the bulk velocity above the crest and the hill height h, was kept at $Re \approx 10^4$.

$$Re = \frac{u_b h}{\nu}; \quad u_b = \frac{1}{2.035h} \int_h^{3.035h} u(y)\mathrm{d}y \tag{1}$$

As the flow features all phenomena of separation from a curved surface and reattachment and the computational cost is relatively low, this configuration has been selected by various researchers, e.g. Temmerman and Leschziner (2001), Jakirlić et al. (2002) and Šarić et al. (2007) as

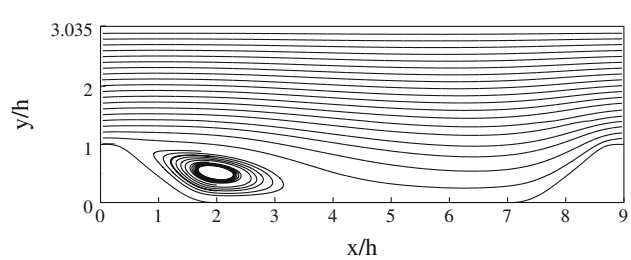

Fig. 1 Streamlines from PIV measurements at $Re \approx 10{,}600$ (Rapp et al. 2008)

[1] Earlier works of Prandtl (1926), von Kármán (1930), Nikuradse (1932) and Nikuradse (1933) dealt with smooth and rough surfaces rather than the definition of wavy walls by Fromm (1923) and Hopf (1923).

[2] Denoted as 'periodicity' is the periodic behavior in the streamwise direction, whereas 'homogeneity' refers to the spanwise direction.

benchmark test for turbulence models and numerical methods. In Fröhlich et al. (2005) detailed flow investigations have been performed based on large eddy simulation (LES). However, at this stage, the flow was only investigated by LES and no reliable experimental data were available.

To fill this lack and to investigate the Reynolds number dependence of the flow, a water channel has been set up in the Hydromechanik Laboratorium of the Technische Universität München. The first experimental results were presented at the TSFP 5 (Rapp and Manhart 2007) and at the DLES 7 (Rapp et al. 2008). Experimental data at $Re = 5{,}600$ and $Re = 10{,}600$ were first compared with DNS and highly resolved LES in a study by Breuer et al. (2009). A detailed study is contained in (Rapp 2009).

Besides the geometry and the coordinate system referred to later, Fig. 1 sketches streamlines from PIV measurements at $Re = 10{,}600$. From these, it can be inferred that the flow is strongly accelerated at the windward slope of the hill, separates from the crest, and forms a large recirculation zone in the wake of the hill.

The asymptotic theory (e.g. Jackson and Hunt 1975; Sykes 1980; Hunt et al. 1988; Belcher et al. (1993) is able to predict velocity speed up over small sinusoidal hills in turbulent boundary layers. In our case, the speed up at the crest of the hill might be determined to leading order by the blockage ratio of about 1/3. There might be a local acceleration departing from that determined by factors such as the ratio of the wave length to the hill height and the local slope and curvature of the hill. Motzfeld (1937) measured velocity profiles over series of hills with different shapes. The local speed up is strongly dependent on the slope of the hills. One configuration produces a near-wall velocity maximum that is even larger than the free stream velocity. The same is reported by Almeida et al. (1993) with a stronger velocity maximum over the single hill which is 27% larger than the free stream velocity which results in a speed-up factor of 2.74 being considerably larger than what was observed in other studies (e.g. Athanassiadou and Castro 2001; Gong and Ibbetson 1989; Mason 1986; Zeman and Jensen 1987).

The point of natural reattachment lies approximately 4.5 hill heights downstream from the hill crest in the middle of the bottom plate which renders the separation zone a little bit shorter than in Almeida et al.'s single hill experiment. A monotonic decrease of the recirculation length was reported by Breuer et al. (2009) for $2{,}800 < Re < 10{,}595$. According to experiences in other flows (Armaly et al. 1983), we expect the recirculation length to settle at some higher Reynolds numbers.

In the course of the paper, we present flow measurements by PIV and LDA up to $Re = 37{,}000$. In a real experiment, no periodic and homogeneous boundary conditions can be met. Therefore, we assess whether the flow can be considered as periodic in streamwise and homogeneous in spanwise direction or if the three-dimensionality affects the whole flow field. We focus on the development of the mean flow and the turbulence structure with the Reynolds number and investigate secondary flow structures at the windward side of the hill. An open point is the development of the velocity speed up over the crest of the hill. It is not evident if the theory developed for small disturbances can be applied in the present flow case. Another issue of this paper is the structure and behavior of the recirculation zone and their Reynolds number dependence.

The paper is structured as follows. After a description of the experimental setup (Sect. 2) and the measurement systems used (Sect. 3), we present and discuss results of our measurements in Sect. 4. A detailed analysis of the homogeneity and periodicity of the flow is undertaken and the flow region influenced by side-wall effects is evaluated. Two-point correlations in the spanwise direction document the size of the main flow structures and their development with the Reynolds number. The time averaged flow field in the symmetry plane of the channel is thoroughly investigated including the mechanism that drives the speed up of the flow above the hill. Second order statistics are presented and discussed as well as the structure and dynamics of the main recirculation zone. A further look is taken into the velocity PDF at the hill top to elucidate possible separation on the hill crest. Finally, second order statistics are used to shed some light on the splatting effect at the windward side of the hill.

2 Experimental setup

A water channel of eleven meters in length has been set up in the Hydromechanik Laboratorium of the Technische Universität München to investigate the flow over periodic hills experimentally. The dimensions referred to in this chapter relate to the hill height h that was chosen to be 50 mm.

2.1 Fluid

We chose water to study the flow as the kinematic viscosity is lower and the density is higher compared to the air's properties. The water temperature changed by about 2°C during the whole measuring period. Therefore, the kinematic viscosity underwent a change of $\Delta\nu \approx 5\%$ according to Poiseuille[3] which was accounted for in the calculation of

[3] $\nu = \frac{1.78\times10^{-6}}{1+3.37\times10^{-2}T+2.21\times10^{-4}T^2}\left[\frac{\mathrm{m}^2}{\mathrm{s}}\right]$, with T [°C].

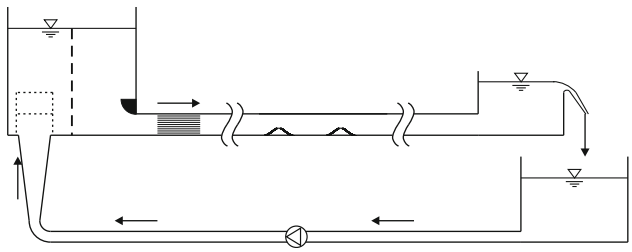

Fig. 2 Experimental setup

the respective Reynolds numbers. The water was filtered, decalcified, and chlorinated. It was necessary to filter the water to remove particles from it, so that one can be sure that the detected particles were those used for seeding.

2.2 Material

The channel including the inlet and outlet reservoirs was realized in 20-mm PVC panels. In the measuring section 1.80-m long plexiglass (thickness 20 mm) forms the channel's top cover to make the flow optically accessible. The sidewalls are made of 8.0 mm optical glass in this area. The hills consist of polyurethane and were cast from monolithic blocks. The surface roughness is ≤ 1 µm. The joints of the channel's panels were filled with epoxy resin and polished.

2.3 Hydraulic system

To reduce influences of probable pump fluctuations on the flow, the channel was divided into two quasi separate hydraulic systems.[4] One consists of the main reservoir, the pipe, and the inlet reservoir with a free surface. The second one begins with the inlet reservoir that is connected to the channel which ends with an outlet reservoir that has got a free surface as well (see Fig. 2). The pump used has got a nominal power of 14 kW providing a maximum discharge of 70 ℓ/s. The discharge is monitored with a magneto-inductive device (Fischer-Porter D10D) and regulated by a slide to adjust the Reynolds number. The water enters a 1.8-m long, 0.80-m high and 0.90-m wide inlet reservoir through a diffusor (see Fig. 3). Several fixtures calm down the entering jet before the water is led into the channel. The water surface oscillations in chamber one are completely eliminated in chamber three (see Fig. 4).

The rectangular channel containing the hills has got a height of 151.75 mm which corresponds to 3.035h. The width of the channel was chosen to be 18 h to achieve homogeneity in the spanwise direction.

[4] However, the discharge was continuously recorded by a magneto-inductive device. The signal was analyzed but fluctuations were not detected in the spectra.

A 2-m long section is placed before the hills' section in which flow straighteners (diameter 22 mm or 0.44h) fill out the whole cross section at a length of 0.50 m or 10h ($l \geq 20 d_r$, Eckelmann 1997), respectively. A distance of 1.0 m or 20h lies between the flow straighteners and the foot of the first hill. The overall entrance length corresponds to 30 hydraulic radii.

In total, ten hills with an inter-hill distance of 0.45 m (9h) follow and constrict the channel from 151.75 (3.035h) to 101.75 mm (2.035h). A rectangular channel (1.70 m, 34 h) was built in between the last hill and the outlet reservoir. The outlet reservoir is 1.0 m (20h) long and ends with a weir over which the water flows back to the main reservoir.

2.4 Seeding particles

Hollow glass spheres with a diameter d_p of (8 µm $\leq d_p \leq$ 12 µm) were chosen to seed the completely particle free water. Their densities range from $\rho_{p(\min)} = 0.1\,\mathrm{g/cm^3}$ to $\rho_{p(\max)} = 1.5\mathrm{g/cm^3}$. To investigate whether these particles follow the flow ideally, the Stokes number is the measure of choice. It is defined as the ratio between the particle time scale $t_p = \frac{\rho_P}{\rho_F}\frac{d^2}{18\nu}$ and the Kolmogorov time scale $t_k = \sqrt{\frac{\nu}{\epsilon}}$.

$$\mathrm{St} = \frac{t_p}{t_k} \tag{2}$$

As long as the Stokes number is smaller than one, we assume that the particles follow the smallest (the Kolmogorov) scales of the flow. The particle relaxation time becomes maximum for the greatest diameter and the greatest density $t_{p(\max)} = 1.20 \times 10^{-5}$ s. The shortest Kolmogorov time scale will be reached in regions of highest dissipation rate ϵ. We estimated the value of the highest dissipation rate from DNS results at $Re = 5{,}600$ (Peller et al. 2006), which gave Kolmogorov time scales using maximum dissipation ($\epsilon_{\max}$) and mean dissipation (ϵ_{mean}) in the flow field, respectively, of $t_{k(\max)} = 4.05 \times 10^{-4}$ s and $t_{k(\mathrm{mean})} = 1.96 \times 10^{-3}$ s. From $\eta \sim Re^{3/4}$ it can be found that the dissipation scales with $u^3/\mathcal{L}$. This yields the maximum and mean Stokes numbers for the Reynolds numbers investigated according to Table 1.

From this table, it can be assumed that the particles follow the flow nearly ideally. The suspended particles were continuously added to the flow via a syringe pump.

3 Measurement setup

We investigated the flow field by 2D PIV and 1D LDA. The PIV measurements were performed in various planes giving insight into the spatial structure of the flow field. The LDA measurements were undertaken to validate the

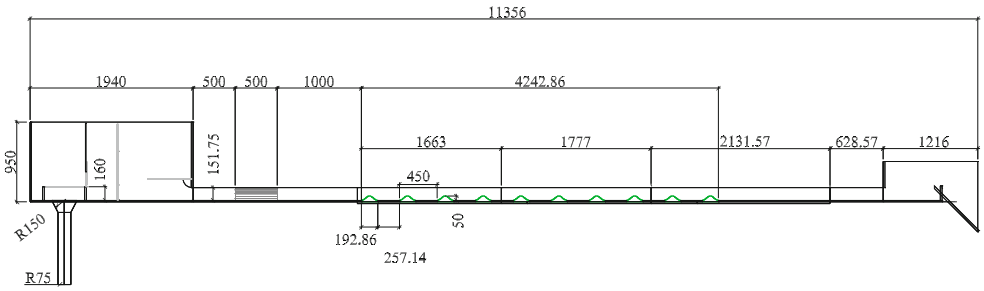

Fig. 3 Longitudinal section of the water channel in [mm]

Fig. 4 Picture of the inlet reservoir at $Re \approx 37{,}000$

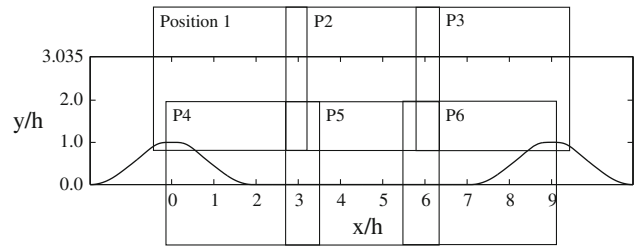

Fig. 5 Combination of six image frames in one measurement plane

Table 1 Maximum and mean Stokes numbers of the seeding particles for the Reynolds numbers investigated

Re	St_{mean}	St_{max}
5,600	6.11×10^{-3}	2.96×10^{-2}
10,600	1.59×10^{-2}	7.72×10^{-2}
19,000	3.82×10^{-2}	1.85×10^{-2}
37,000	1.04×10^{-1}	5.03×10^{-1}

Table 2 Resolution of the interrogation areas with respect to the minimum and mean Kolmogorov scale η

Re	$x/h = 0.5$		$x/h = 1.0$		$x/h = 6.0$	
5,600	29.4	17.2	26.8	17.5	23.4	16.0
10,600	47.5	27.8	43.3	28.30	37.7	25.8
19,000	73.5	43.0	67.0	43.8	58.4	40.0
37,000	121.2	70.9	110.5	72.2	96.3	65.9

PIV algorithm and to get reliable flow information close to the wall.

3.1 PIV measurements

Our 2D PIV system consisted of two pulsed 190 mJ Nd:YAG lasers, a lightarm with light sheet optics, a 4 Mpx CCD camera, a synchronizer and a PC with optical drives. These drives can store $\approx$10,000 double frames at a frame rate of 7.25 Hz.

3.1.1 Vertical light sheets

The light sheet optic was mounted on a 2D traverse above the measurement area. The laser beam was expanded via two concave lenses such that it covered an area of 45 cm ranging from one to the next hill.

A convex lens bundled the light sheet such that its waist ($\approx$0.3 mm) was within the water channel.

Laser light reflections on the bounding walls were avoided by adjusting the camera in the particular optical axis of the system. To cover the whole area between two consecutive hills with overlapping images at a chosen resolution of 0.141 mm/px six camera positions were necessary (see Fig. 5). With a size of an interrogation area of 32 px, the spatial resolution is $0.09h$. Based on Kolmogorov scale η the resolution ranges from about 16η at $Re = 5{,}600$ to 120η at $Re = 37{,}000$, see Table 2.

The camera was mounted on a 3D traverse frame such that it could be positioned with an accuracy of 0.3 mm in each direction. The position of the image could be determined with an accuracy smaller than one pixel in the image plane which corresponds to a positioning error of $\varepsilon < 0.141\,\text{mm} = 2.82 \times 10^{-3}\,h$.

Due to the fragmentation into six individual measurement areas, an experiment consists of six particular measurements with similar but not identical conditions. Therefore, instantaneous information of the whole flow field is not available. In addition, the measurement results had to be normalized. To do so, we have chosen the bulk

Table 3 Nominal and individual Reynolds numbers of PIV measurements

$Re = 5{,}600$, hill 6–7			$Re = 5{,}600$, hill 7–8		
Re_{min}	Re_{mean}	Re_{max}	Re_{min}	Re_{mean}	Re_{max}
5,093	5,186	5,245	5,036	5,119	5,196
$Re = 10{,}600$, hill 6–7			$Re = 10{,}600$, hill 7–8		
Re_{min}	Re_{mean}	Re_{max}	Re_{min}	Re_{mean}	Re_{max}
9,694	9,758	9,812	9,952	10,063	10,153
$Re = 19{,}000$, hill 6–7			$Re = 19{,}000$, hill 7–8		
Re_{min}	Re_{mean}	Re_{max}	Re_{min}	Re_{mean}	Re_{max}
			18,950	19,239	19,396
$Re = 37{,}000$, hill 6–7			$Re = 37{,}000$, hill 7–8		
Re_{min}	Re_{mean}	Re_{max}	Re_{min}	Re_{mean}	Re_{max}
			36,707	36,847	37,099

velocity u_b at $x/h = 0$ from frame $P1$ as a reference. The normalization factor for the other positions is obtained by an integration of the ensemble averaged u-component $\langle u \rangle$ at several x/h positions in the areas in which the frames are overlapping. The ratio of the two mean velocities in the overlapping area then leads to the factor with which the velocities in the other frames were multiplied. The actual Reynolds number of the particular experiment was determined with u_b and the temperature-dependent kinematic viscosity ν according to Sect. 2.1. Table 3 shows the smallest, largest, and the mean Reynolds number of the six particular frames.

3.1.2 Horizontal laser sheets

To investigate the homogeneity of the flow, measurements were conducted in horizontal planes. The camera was mounted above the channel. One measurement series was conducted with a lens with a focal length of $\mathcal{F} = 50$ mm at $Re = 5{,}600$, $Re = 10{,}600$, $Re = 19{,}000$ and $Re = 37{,}000$ with a light sheet position of $y/h = 1.53$. The resolution was similar to the one used for the vertical planes (0.1405 mm/px). These planes reached from approximately $x/h = 2.0$ to $x/h = 7.0$. The extent in the z direction lay between $z/h = -5$ and $z/h = 0$, thus on the left channel half.

Additional experiments were conducted with a wide-angle lens $\mathcal{F} = 24$ mm at $Re = 5{,}600$, $Re = 10{,}600$, $Re = 19{,}000$. For these experiments, two vertical positions were chosen for the light sheets $y/h = 1.53$ and $y/h = 0.40$. With this lens, the measurement area could be extended from the center of the channel $z/h = 0$ to the left side wall $z/h = -9$. The resolution for the upper position was 0.306 mm/px, whereas the lower light sheet led to 0.320 mm/px. The upper images reach from ($x/h = 2$ to $x/h = 7$); the lower ones are delimitated by the hills at $x/h = 1.06$ and $x/h = 7.94$.

3.1.3 Image processing

The PIV sequences consist of 10^4 double images at a frame rate of 7.25 Hz. In a first step, the pictures were clipped and the minimum intensity of each pixel was subtracted (Fore et al. 2005; Keane and Adrian 1990). The interrogation areas covered 32 px at an overlap of half their side length $\mathrm{IA}_L/2$. The evaluation of the images was done in Fourier space with an adaptive cross correlation algorithm in *Insight 3G* from *TSI* which uses two iterations.

In a first iteration loop, a preliminary velocity is estimated that is used as initial shift of the second frame in the second iteration step. Vectors in the first iteration step are assumed to be valid if (1) the signal-to-noise ratio (1st to 2nd peak) of the images was above 1.5, (2) it was within the range $-u_b \le u \le 2u_b$; $-u_b \le v \le u_b$, and (3) it passed a median filter (Westerweel 1994). The outliers ($\approx 5\%$) were interpolated by their surrounding neighbors. In the second cross correlation step, the vectors were validated upon the ratio of first-to-second correlation peaks and global high and low pass filters. The ratio of the outliers fell below 3%. The displacements were sub-pixel-refined with a Gaussian 3-point estimation (Raffel et al. 1998).

Afterward, the vectors are normalized according to their positions as explained above and divided by u_b. For computing u_b the resolution at the wall was refined with a specific wall model otherwise u_b is underestimated because the viscous region was not resolved. The near-wall profile was approximated by the law-of-the-wall (Eq. 3) (Schlichting 1964).

Its range of application is given in terms of wall units ($y^+ = \frac{u^* y}{\nu}$) with $u^* = \sqrt{\tau_w/\rho}$. The near-wall region thus was approximated linearly ($u/u^* = y^+$) in the viscous layer ($y^+ \le 11$) and in the range from $y^+ = 11$ up to the first measurement point ($35 \lessapprox y^+ \gtrapprox 200$) with the logarithmic wall law with a von Kármán constant $\kappa = 0.4$.

$$\frac{u}{u^*} = \frac{1}{\kappa} \ln\left(\frac{y u^*}{\nu}\right) + 5.5 \tag{3}$$

Although the preconditions for applying the logarithmic law are not fully kept, the velocity profile has to be approximated at the upper wall when comparing the velocity profiles with simulations. It can be subsumed that u_b increased by approx. 1–2% when applying the logarithmic law instead of a simple linear law.

Table 4 Flow-through times

Re	t_{fl}
10,600	570
19,000	1,000
37,000	2,000

3.1.4 Statistics

The PIV measurements were done at a frame rate of 7.25 Hz; thus, for each experiment, a total of 10^4 double images were captured. This results in a measuring time of $t_M = 1{,}380$ s or 23 m, respectively, and in the flow-through times t_{fl}

$$t_{fl} = \frac{u_b t_M}{9h} \tag{4}$$

as indicated in Table 4. The minimum flow-through time is $t_{fl} = 570$ for the lowest Reynolds number, which is far higher than what in LES of that flow case was reported Fröhlich et al. (2005) and Breuer et al. (2009). It should be highly sufficient to resolve the low-frequency oscillations of the separation bubble that scale with the flow-through time.

3.2 Measurement setup LDA

The LDA system that was used is a cross-beam system (Fig. 6). It consists of a diode pumped Nd:YAG laser emitting light at a wavelength of $\lambda = 532$ nm with a power of 75 mW. The frequency of one beam is shifted by a Bragg cell such that reverse flow can be determined. The scattered light is led by a fiber cable to the photomultiplier. The signal was filtered analogously, amplified, and digitized at a rate of $f_{a(\max)} = 50$ MHz. The laser was mounted on a 3D traverse system. The measurement volume was $b_{MV} = 2.52 \times 10^{-4}$ m in the streamwise and the vertical directions and $l_{MV} = 8.96 \times 10^{-3}$ m in the spanwise direction.

The measurement accuracy was determined by the vendor as smaller than 1%. The settings of the LDA system were kept constant throughout the experiments. The frequency-shifted signal was amplified by a factor of 10 and highpass (300 kHz) and lowpass filtered (3 MHz). The sampling frequency was 10 MHz, the shift frequency 1 MHz (Nyquist Theorem $f_{\text{sampl.}} > 2 \cdot f_{\text{shift}}$). The trigger level was set to 10^{-4}mV, before $N = 4{,}096$ samples of the signal were Fourier-transformed. The discrete velocity resolution refined by a parabolic fit was $\Delta u = \frac{\Delta f \lambda}{2 \sin \phi} = 0.0173$ m/s. The LDA measurements were conducted at four streamwise positions. Twenty measurement points were chosen at $x/h = 0.05$ and 29 at $x/h = 4.0$. The equidistant distance between the measurement points correspond to the size of an interrogation window of the PIV measurements. As the velocity is measured at a time and a spot the profiles obtained are rougher than the ones from the PIV. The measurements at each point were conducted periodically to reduce low-frequency impacts of the flow. All in all 25 measurements were done at every point for 10, 30 und 60 s with a total measuring time of 15 min which resulted in 35,000 to 45,000 samples per point.

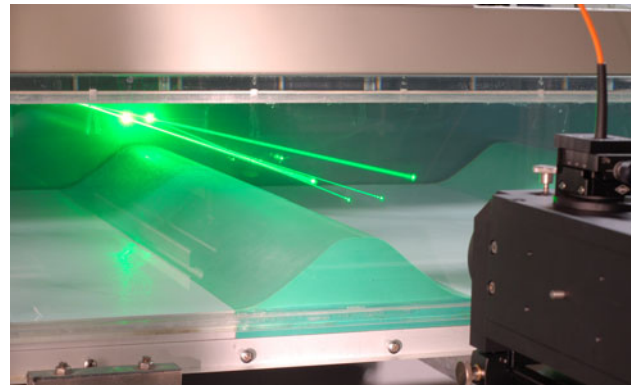

Fig. 6 Picture of the LDA system

4 Results

This section comprises the validation of the PIV measurements, the control of periodicity in the streamwise and homogeneity in the spanwise direction as well as investigations into certain flow features.

4.1 Validation

To strengthen the reliability of the PIV data, LDA measurements were conducted at $x/h = 0.05$ and $x/h = 4.0$ in the center of the channel for various Reynolds numbers. The former position is close to the point of separation, and the latter is in the area of mean reattachment. In this section, the measurement results are counter-checked with each other. Figures 7 and 8 show average velocity profiles and streamwise turbulence intensities at the latter position ($x/h = 4.0$) at $Re = 10{,}600$. While the mean streamwise velocity profiles comply perfectly between the two measurement systems, minor deviations were detected in the rms values of the u component obtained by PIV and LDA. The LDA values are slightly smaller at this position than the respective PIV ones. This minor offset results from the determination of u_b which has got a quadratic impact on the normalization and is therefore sensitive. However, the agreement is fully satisfying.

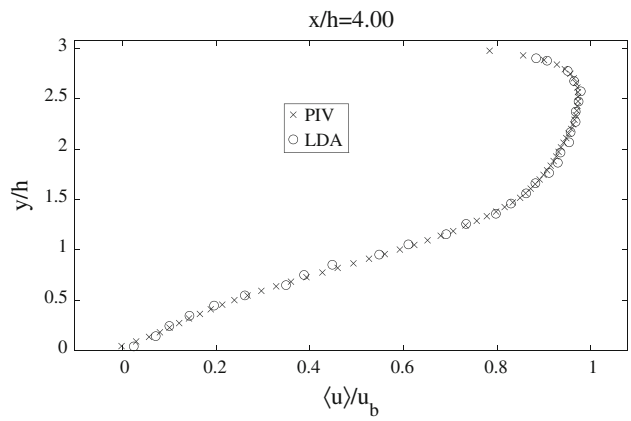

Fig. 7 Comparison of $\langle u \rangle / u_b$ from PIV and LDA measurements $Re = 10{,}600$; $x/h = 4.0$

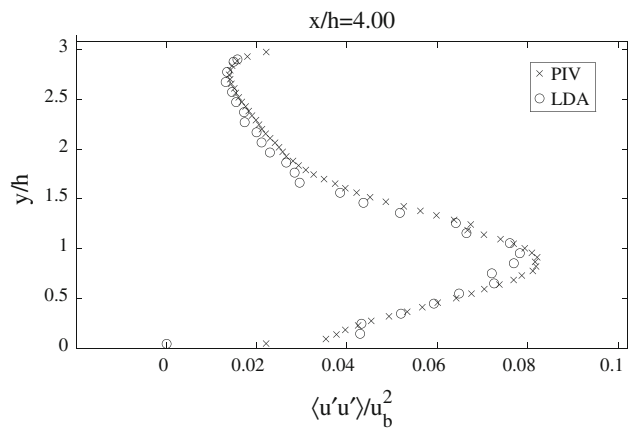

Fig. 8 Comparison of $\langle u'u' \rangle / u_b^2$ from PIV and LDA measurements $Re = 10{,}600$; $x/h = 4.0$

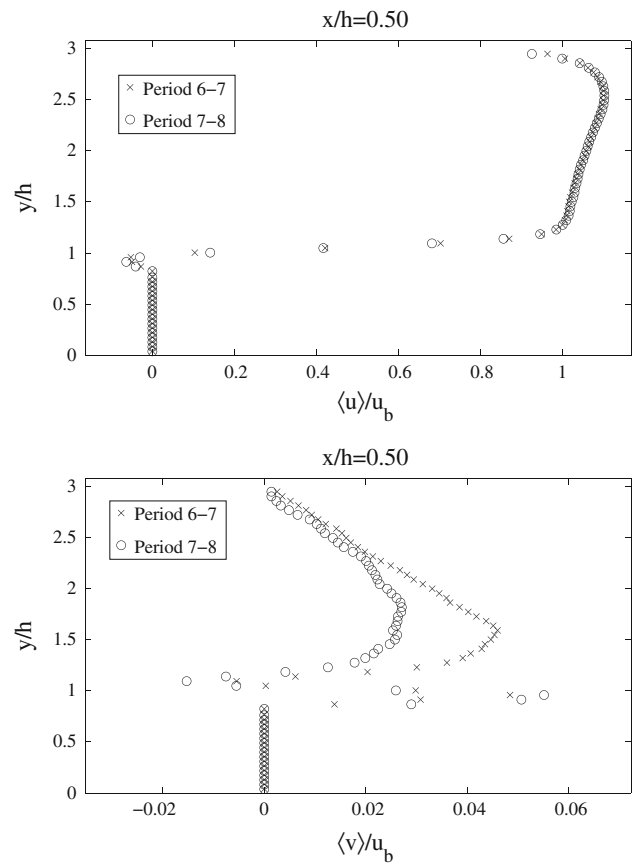

Fig. 9 Comparison of velocities $\langle u \rangle / u_b$ and $\langle v \rangle / u_b$ between hill pairs 6–7 and 7–8 at $x/h = 0.5$; $Re = 5{,}600$

The complementary operation of PIV and LDA measurements enhances the preferences of the particular technique. With the data obtained, it can be concluded that the PIV data are reliable so that the valuable field information of the PIV measurements can be used as reference.

4.2 Periodicity in the streamwise direction

One of the goals of the experimental setup was to achieve periodic flow conditions in the streamwise direction to exclude influences from upstream disturbances on the flow. To achieve this, ten hills were placed in the channel in total. Henn and Sykes (1999) investigated the influence of the number of periods on the flow over sinusoidal waves numerically. They found crucial differences by comparing the results of one and two periods at a Reynolds number of $Re \approx 10^4$, whereas only minor deviations appeared for two and four periods. Additionally they conducted measurements in a water channel at the eleventh and twelfth of 16 wave trains. They did not find noteworthy differences in the average values of the hot-wire and cross-wire measurements. Similarly, Zilker et al. (1977) report for a similar setup as considered here, periodicity was obtained after six waves.

We certainly expect that the periodicity of the flow will be subject to Reynolds number influences. To investigate the periodicity of the flow, we compare PIV measurements between the hill pairs 6–7 and 7–8 at $Re = 5{,}600$ and 10,600. Additionally LDA measurements were conducted at $Re = 10{,}600, 19{,}000$ and 37,000 at the positions $x/h = 0.05$ and 4.0 between the same hill pairs.

Re = 5,600 Figures 9 and 10 depict the results from PIV measurements at two consecutive periods at $Re = 5{,}600$. They compare $\langle u \rangle / u_b$ and $\langle v \rangle / u_b$ at $x/h = 0.5$, i.e. at the hill crest, and $x/h = 4.0$ close to reattachment.

A good compliance of the u components is found in Fig. 9 at $x/h = 0.5$. However at $x/h = 4.0$ (Fig. 10) there are slight deviations. The profile between hill 7 and 8 is flatter which goes in hand with a shorter reattachment point. The flow reattaches at $x/h = 4.83$ between hill 7 and 8 whereas between hill 6 and 7, it reattaches at $x/h = 5.02$. The discrepancy becomes clear when looking at the v components in Fig. 9. Between hills six and seven the

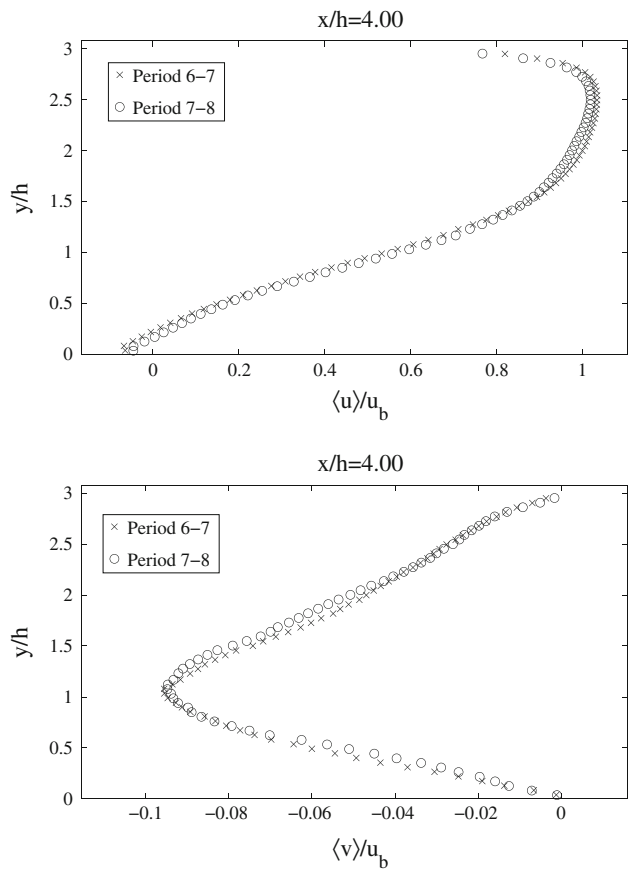

Fig. 10 Comparison of velocities $\langle u\rangle/u_b$ and $\langle v\rangle/u_b$ between hill pairs 6–7 and 7–8 at $x/h = 4.0$; $Re = 5{,}600$

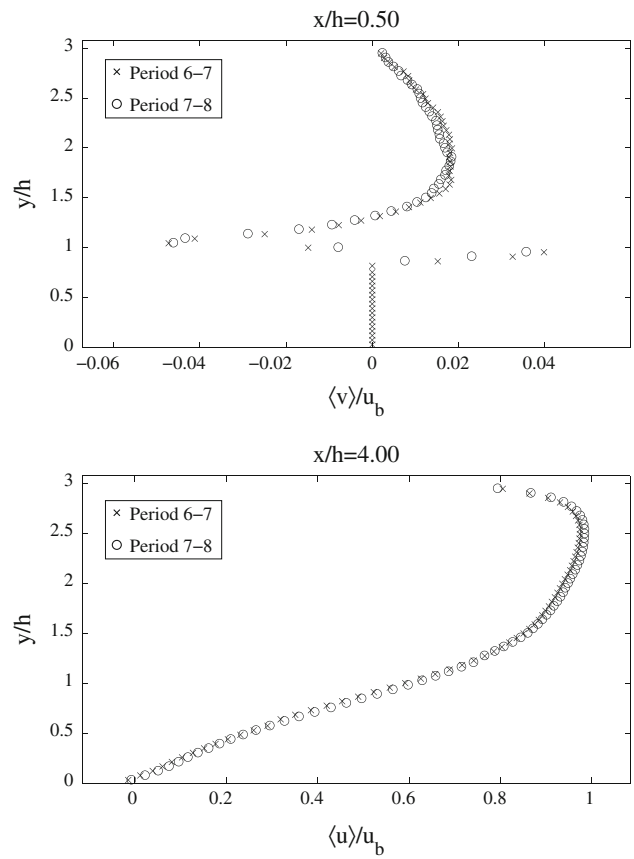

Fig. 11 Comparison of velocities $\langle v\rangle/u_b$, $\langle u\rangle/u_b$ between hill pairs 6–7 and 7–8; $Re = 10{,}600$

vertical component at $x/h = 0.5$ is greater, which suggests a later reattachment. The greatest discrepancies in the whole flow field can be found in the shear layer at $x/h = 2.0$; $y/h = 0.74$ where the u component differs around 7.0% whereas the corresponding percentage for the v component is 3.5 (not shown).

The rms values of the two periods do not deviate significantly. The periodicity of the flow could not be fully proofed for this Reynolds number. The main indicators of non-periodicity at this Reynolds number were obvious in the profiles of the vertical velocity component at $x/h = 0.5$ and the one of the streamwise velocity component at $x/h = 4.0$. Therefore, these profiles are inspected at the higher Reynolds numbers.

Re = 10,600 Plot 11 shows the profiles of the vertical velocity component at $x/h = 0.5$ and the one of the streamwise velocity component at $x/h = 4.0$. These were the profiles in which the non-periodicity at the lower Reynolds number was obvious. The compliance of the mean velocities is more than satisfying. Only marginal deviations can be found in the vertical component at $x/h = 0.5$. One has to bear in mind that the vertical component is a magnitude smaller than the streamwise one. The mean point of reattachment between hills six and seven lies at $x/h = 4.28$ whereas the flow reattaches at $x/h = 4.21$ between hills seven and eight.[5] The difference of $0.07h$ is minor; however, the LDA data support these findings. The LDA measurements at $x/h = 4.0$ at the wall nearest measurement point lead to $\langle u\rangle/u_b = 0.0223$ for the period 6–7 and ($\langle u\rangle/u_b = 0.0242$) for 7–8. Hence, the flow is recovering earlier at the latter, which complies with the PIV data (Fig. 11).[6]

The greatest difference in the mean streamwise component between two corresponding points in the two periods are found at $x/h = 4.0$ and $y/h = 1.1$ and can be quantified to $\Delta u = 1.2\%u_b$. The greatest deviation of the vertical component was found in the shear layer at $x/h = 3.0$ and $y/h = 0.55$ and is $\Delta v = 0.8\%u_b$. According to these observations, one can conclude that the flow is periodic at $Re = 10{,}600$.

Re = 19,000 After the periodic behavior of the flow was shown for $Re = 10{,}600$, it can be assumed that for higher

[5] The method of determining the point of reattachment will be explained in Sect. 4.7.

[6] Temmerman (2004) compared LES with one and two periods and found differences in the point of reattachment of 0.05 hill heights.

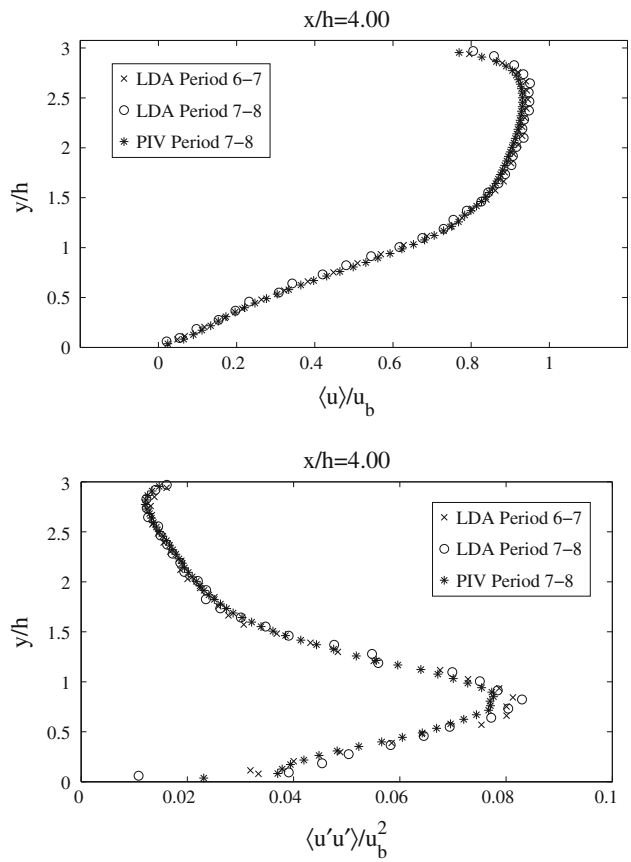

Fig. 12 Comparison of $\langle u\rangle/u_b, \langle u'u'\rangle/u_b^2$ from PIV and LDA measurements between hill pairs 6−7 and 7−8 $x/h = 4.0$; $Re = 19{,}000$

Reynolds numbers, the flow periodicity is even more likely to occur. However, LDA measurements were carried out at $Re = 19{,}000$ at the sensitive position $x/h = 4.0$. Figure 12 illustrates the agreement of the mean and the rms values of the two periods analyzed. The variations at the peak probably result from the finer LDA resolution (greater measurement volumes smooth the fluctuations).

The same observations hold for the normalized Reynolds stresses too. Discrepancies can only be found in the range of the measurement accuracy.

To summarize, the PIV and LDA measurements at hill pairs 6–7 comply for $Re \geq 10{,}600$ very well with the ones from hills 7–8 so that a periodic behavior can be assumed for Reynolds numbers greater or equal 10,600. Further experiments at $Re = 37{,}000$ that will not be discussed in this section support this conclusion.

4.3 Homogeneity in the spanwise direction

The experiment was designed to achieve a two-dimensional, i.e. homogeneous, flow in the center of the channel in order to study a 2D flow situation and to exclude 3D effects on the flow statistics. While most of the studies on periodic or single hills cited here claim two-dimensionality, only in some of them it was demonstrated explicitly that three-dimensional effects were absent.

The spanwise width of the experiment has to be large enough that the influence of the side walls vanishes. This is a function of spanwise/vertical aspect ratio. Of course, perfect two-dimensionality cannot be achieved in an experiment with side walls. If three-dimensional effects can be neglected in the respective governing equations, however, we assume that the terms homogeneity or two-dimensionality would be applicable for such a situation.

From the numerical simulations (Fröhlich et al. 2005), we know that two-point correlation lengths indicate spanwise structures larger than $4.5h$ but presumably smaller than 7–8. Therefore, the width of the channel should cover at least two times that correlation length to have a flow in the middle of the channel that is independent of the side walls. Most of the LES used periodic domains of $4.5h$ in the spanwise direction (e.g. Mellen et al. 2000; Fröhlich et al. 2005; Temmerman and Leschziner 2001; Temmerman et al. 2003; Peller and Manhart 2005; Breuer et al. 2009). The issue of the lateral width arises if results shall be compared between experiment and numerical simulations. Although the experimental domain cannot fulfill periodic boundary conditions, it will be shown that it is wide enough to cover even the largest structures expected in the flow.

A channel width of $18h$ was realized in the present experiment in order to achieve homogeneous or 2D flow conditions, respectively, in the center of the channel. Despite this large aspect ratio, homogeneity could be violated due to two effects, (1) influence of the side walls and (2) secondary flow structures in distinct flow regions, such as Taylor–Görtler vortices. In the following, mean and rms values of the u and w velocity components and individual terms of the momentum balance are analyzed in terms of homogeneity in the spanwise direction.

PIV measurements with horizontal laser light sheets were accomplished with a resolution of 0.14 mm/px at a height of $y/h = 1.53$ in the area of $2.5 \lessapprox x/h \lessapprox 7$ and $-5 \lessapprox z/h \lessapprox 0$. Further experiments at this height and at $y/h = 0.40$ were carried out with a coarser resolution (0.31 mm/px) such that the flow could be investigated from the center to the side wall.[7].

The velocity profiles of the horizontal PIV planes are rougher than the ones of the vertical planes. This has got strong impacts on the gradients that will be discussed in the course of this section. To reduce such effects, the velocity

[7] The measurements were only conducted at the left channel half such that genuine symmetry about the centerline has not been demonstrated.

fields were Gaussian-filtered in the horizontal and the streamwise direction according to Eq. (5).

$$G(x) = \frac{1}{\Delta + 1}\sqrt{\frac{6}{\pi}}\exp\left(-\frac{6x^2}{\Delta^2}\right) \tag{5}$$

A filter width of $\Delta = 8\mathrm{IA}_L$, with the length of an interrogation area IA_L, was used.

4.3.1 Secondary flow structures

Secondary flow structures such as Taylor–Görtler vortices could arise from the unstable curvature of the streamlines on the windward side of the hill. Such structures have been observed on the windward side of nominal two-dimensional flow obstacles, thus destroying the two-dimensional flow conditions (Dimaczek et al. 1989). In numerical simulations employing periodic boundary conditions in the spanwise direction, such as the one by Fröhlich et al. (2005), the structures are not fixed in space and can fluctuate freely in the spanwise direction. Thus, it is not clear whether they could have been detected in a LES using periodic boundary conditions. In a real experiment, such vortices would be fixed in the spanwise position due to the blocking effect of the side walls. In order to clarify this issue, the occurrence and amplitude of such flow structures will be documented here.

Figure 13 depicts the mean velocity across the left channel half at $y/h = 0.4$ in a contour plot for $Re = 5{,}600$ and $Re = 10{,}600$ overlaid with vectors of the secondary flow $u_s(z) = u(z) - \bar{u}$, with $\bar{u}$ being the spanwise average of the velocity. u_b was taken from the respective vertical PIV planes.

There is a clear secondary flow structure at $Re = 5{,}600$ which consists of a large-scale vortex with a wave length of about $5h$ in the spanwise direction. Considering a spanwise domain of $18h$, this would imply four secondary vortices placed in the spanwise direction. Their amplitude is not negligible at $Re = 5{,}600$ and leads to a nearly sinusoidal velocity variation of about $5\%u_b$ at $x/h = 7$. At $Re = 10{,}600$, however, only slight variations in the mean streamwise velocity component occur (Fig. 13, bottom). The variations in the streamwise velocity component are less than $2\%u_b$, the ones of the spanwise component are less than $1\%u_b$ and can hardly be identified in this plot. From this comparison, it can be concluded that at the higher Reynolds numbers, the secondary flow structures can be neglected, while they are of considerable impact on the flow field at $Re = 5{,}600$.

4.3.2 Influence of the side walls

In order to quantify the range of influence of the side walls, we conducted measurements in two planes, at $y/h = 0.40$

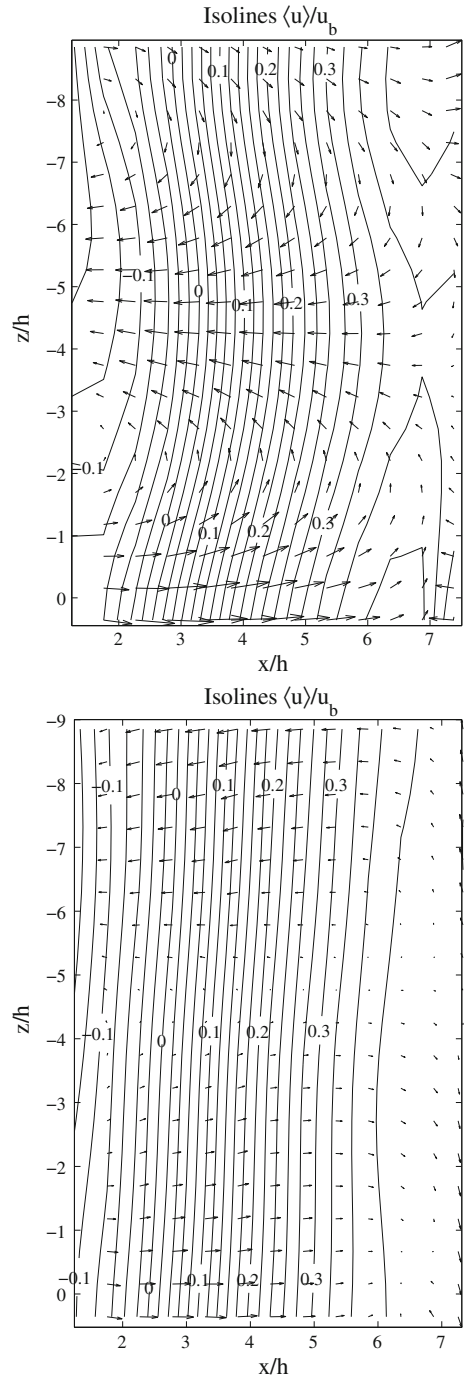

Fig. 13 Isolines of $\langle u\rangle/u_b$ and vectors of mean secondary flow field at $y/h = 0.4$; $Re = 5{,}600$ (*top*), $Re = 10{,}600$ (*bottom*)

and $y/h = 1.53$, respectively. The results shown herein cover the lower plane measurements. Significant variations of the u and the w component could not be determined. But the effects discussed above can be found here too, see Fig. 14. At $Re = 10{,}600$ maximum variations in u are about $2\%u_b$ excluding the near-wall region. This value though has decreased to $1.2\%u_b$ for 19,000. Furthermore,

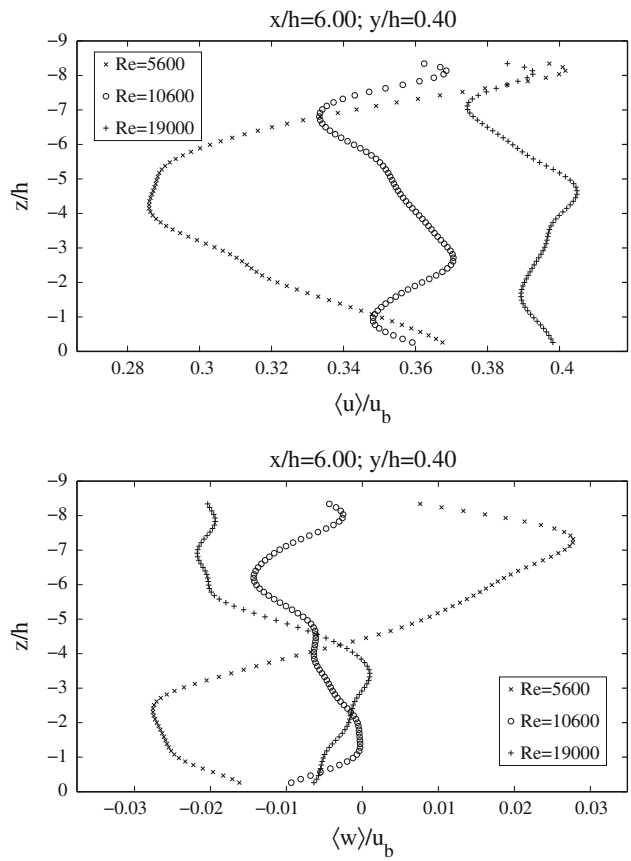

Fig. 14 Lateral variations of $\langle u\rangle/u_b$ (*top*) and $\langle w\rangle/u_b$ at $x/h = 6.0$; $y/h = 0.40$; at different *Re*

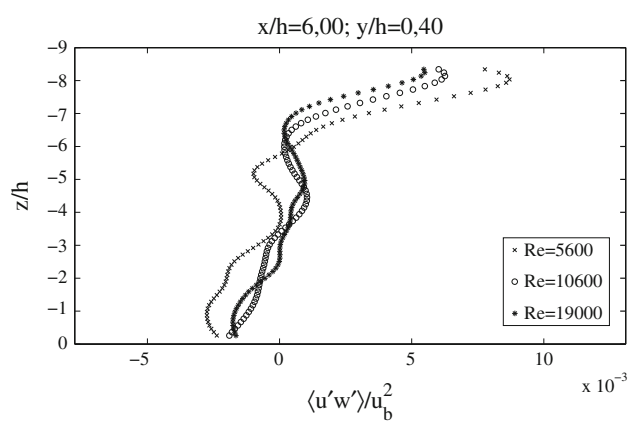

Fig. 15 Lateral variation of $\langle u'w'\rangle/u_b^2$ at $x/h = 6.0$; $y/h = 0.40$; various *Re*

the mean velocity component in the z direction in this plane oscillates around zero but never exceeds a magnitude of $-1\%u_b$.

The greatest impact of the side walls can be observed in the spanwise Reynolds shear stress $\langle u'w'\rangle$ (Fig. 15). Although its magnitude is generally small, it peaks at the side wall ($z/h = \mathrm{f}{-}9$) as expected due to the turbulence production there. It drops to zero at 4 and $3h$ from the wall for $Re = 5{,}600$ and $Re = 19{,}000$, respectively, leaving a domain of $12h$ unaffected by the lateral side wall turbulence generation mechanism.

4.3.3 Consideration of the momentum balance

In a two-dimensional flow, gradients into one direction have to vanish. However, this is not a sufficient condition for two-dimensionality as e.g. a symmetric profile also has one point of zero gradients. We check this issue in our flow field by examining the momentum balance in the center of the channel.

In a physical experiment, it is not possible to acquire the velocity and the pressure field at one instant of time. In addition, the two-dimensional PIV technique that was used limits the number of attainable terms further, such that for the vertical PIV planes gradients in the $x{-}y$ and for the horizontal ones gradients in the $x{-}z$ plane can be estimated. However, at the crossing of the vertical and the horizontal PIV planes, all the terms of the Reynolds equation in the x direction (except for the pressure gradient) can be obtained. Here, we consider the streamwise momentum balance, given by Eq. (6).

$$\langle u\rangle\frac{\partial\langle u\rangle}{\partial x}+\langle v\rangle\frac{\partial\langle u\rangle}{\partial y}+\langle w\rangle\frac{\partial\langle u\rangle}{\partial z}=-\frac{1}{\rho}\frac{\partial\langle p\rangle}{\partial x}+\nu\left(\frac{\partial^2\langle u\rangle}{\partial x^2}+\frac{\partial^2\langle u\rangle}{\partial y^2}+\frac{\partial^2\langle u\rangle}{\partial z^2}\right)-\frac{\partial\langle u'u'\rangle}{\partial x}-\frac{\partial\langle u'v'\rangle}{\partial y}-\frac{\partial\langle u'w'\rangle}{\partial z} \tag{6}$$

In this equation for the x momentum, the following terms should vanish in a nominally two-dimensional flow:

$$\langle w\rangle\frac{\partial\langle u\rangle}{\partial z};\ \nu\frac{\partial^2\langle u\rangle}{\partial z^2};\ \frac{\partial\langle u'w'\rangle}{\partial z}.$$

If these terms are negligibly small, the flow can approximately be considered as two-dimensional in the center of the channel.

In the following, the orders of magnitude of the individual terms will be examined. Additionally, the terms that were obtained from both the vertical and the horizontal planes will be cross-checked with each other. The partial derivatives were computed with a second-order central differencing scheme after smoothing with the Gaussian filter as explained above.

Comparison of the double terms At the intersection line of the vertical measuring plane with the horizontal one at $z/h = 0$ and $y/h = 1.53$, the following terms can be cross-checked $\langle u\rangle\frac{\partial\langle u\rangle}{\partial x}$, $\frac{\partial\langle u'u'\rangle}{\partial x}$ and $\frac{\partial^2\langle u\rangle}{\partial x^2}$ as they are computed from

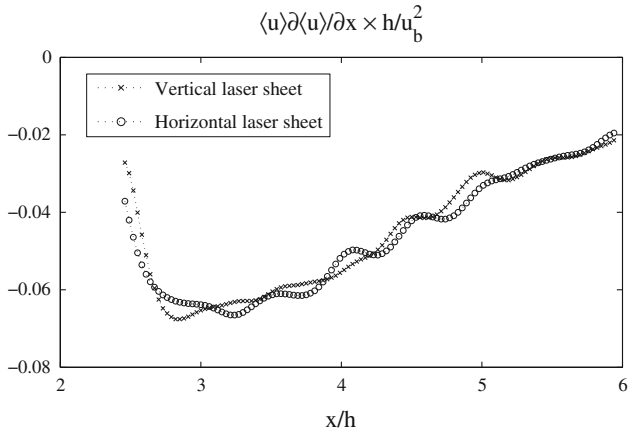

Fig. 16 Comparison of $\langle u\rangle\frac{\partial\langle u\rangle}{\partial x}\cdot\frac{h}{u_b^2}$ from *horizontal* and *vertical* PIV measurements; $y/h = 1.53; Re = 10{,}600$

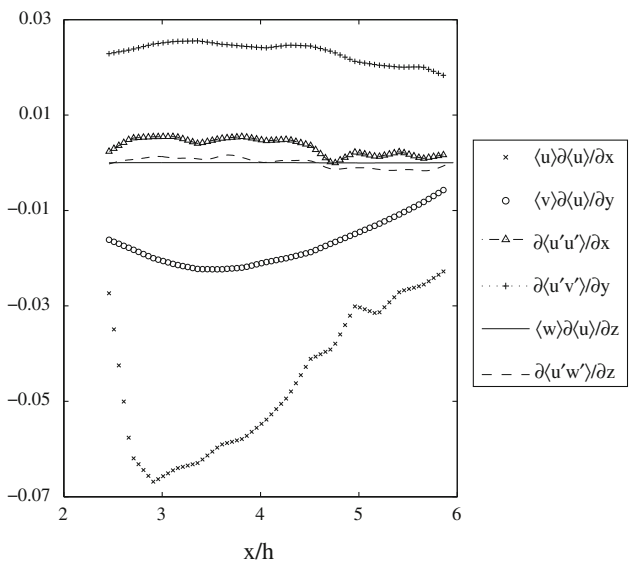

Fig. 17 Normalized terms of the streamwise momentum balance; $y/h = 1.53; z/h = 0.0; Re = 10{,}600$

$x/h \approx 2.5$ to $x/h \approx 6.5$ from two completely independent experiments. As an example, the streamwise convective terms are shown in Fig. 16 for $Re = 10{,}600$ along the x axis (normalized by u_b^2/h).

Despite the fact that the velocity fields were filtered in advance, the gradients show a wiggle-like behavior. Undoubtedly, the tendency of these terms agrees in a satisfactory manner.

Magnitudes of the different terms of the momentum balance Figure 17 depicts the magnitudes of the particular terms of the momentum balance in the x direction at $Re = 10,600$. The viscous term $\nu\frac{\partial^2\langle u\rangle}{\partial x^2}$ is orders of magnitude smaller than the other terms and therefore not shown

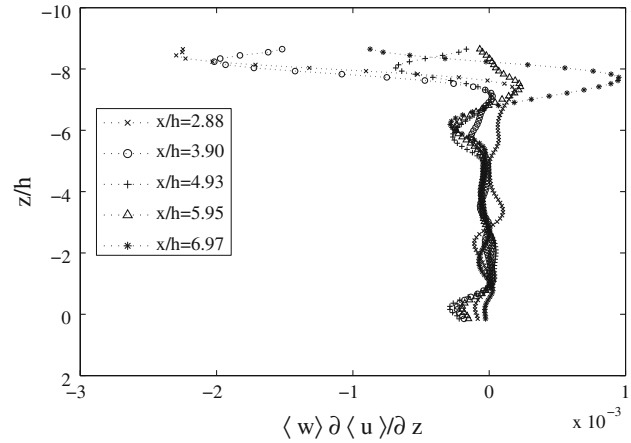

Fig. 18 Convective term $\langle w\rangle\frac{\partial\langle u\rangle}{\partial z}$ across the *left* channel half; $y/h = 0.4; Re = 10{,}600$

here. The dominating terms are the convective terms $\langle u\rangle\frac{\partial\langle u\rangle}{\partial x}$ and $\langle v\rangle\frac{\partial\langle u\rangle}{\partial y}$ as well as the Reynolds term $\frac{\partial\langle u'v'\rangle}{\partial y}$. The greatest contribution of the lateral movement to the momentum balance in the x direction originates from $\frac{\partial\langle u'w'\rangle}{\partial z}$. The value varies from $-1.66\times10^{-3}\cdot u_b^2/h$ to $1.64\times10^{-3}\cdot u_b^2/h$. However, the maximum value of this term is more than one order of magnitude smaller than $\frac{\partial\langle u'v'\rangle}{\partial y}$. The convective term $\langle w\rangle\frac{\partial\langle u\rangle}{\partial z}$ lies between -3.14 and $4.18\times10^{-5}\cdot u_b^2/h$ and is another two orders of magnitude smaller and therefore negligible. The viscous stresses contribute expectedly the smallest part to the momentum balance. E. g. $\nu\frac{\partial^2\langle u\rangle}{\partial x^2}$ varies from $-1.80\times10^{-5}\cdot u_b^2/h$ to $5.98\times10^{-6}\cdot u_b^2/h$. The results for $y/h = 0.4$ behave accordingly.

The same observations were made at $Re = 19{,}000$. The largest three-dimensional impact on the flow results from $\partial\langle u'w'\rangle/\partial z$ which, however, is two orders of magnitude smaller than the 2D terms.

The convective term $\langle w\rangle\frac{\partial\langle u\rangle}{\partial z}$ and the Reynolds term $\frac{\partial\langle u'w'\rangle}{\partial z}$ shall be considered in the following for both Reynolds numbers. These gradients in the z direction contribute most to the streamwise momentum balance. Their profiles in the spanwise direction are plotted in Figs. 18, 19, 20 and 21 for various streamwise positions. First, it has to be noted that they are at least one order of magnitude smaller than the largest terms in the streamwise momentum balance. However, the following observations can be made. Both Reynolds and convective terms peak at the side walls. These peaks extend until maximally $4h$ into the channel leaving a domain of approximately $10h$ unaffected by the distinct side wall effects.

The remaining inhomogeneities that were measured are likely to result from insufficient statistics that becomes visible only in the spatial gradients. These inhomogeneities

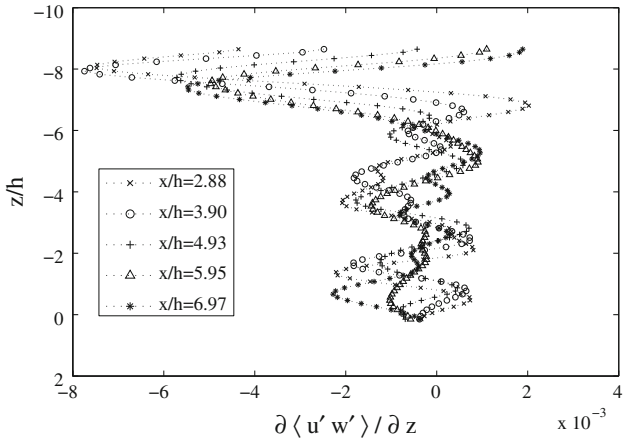

Fig. 19 Reynolds term $\frac{\partial\langle u'w'\rangle}{\partial z}$ across the *left* channel half; $y/h = 0.4; Re = 10{,}600$

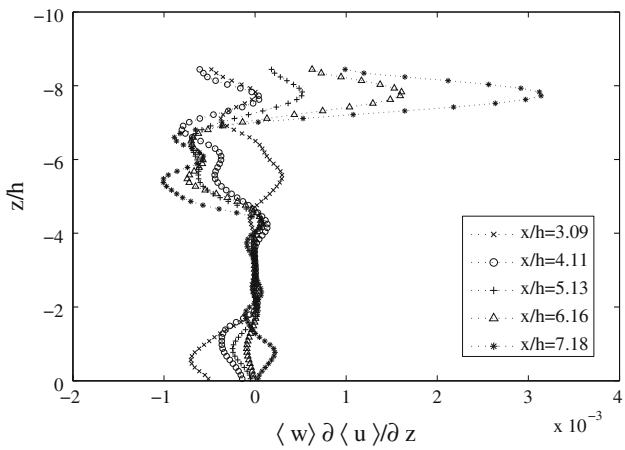

Fig. 20 Convective term across the *left* channel half; $y/h = 0.4$; $Re = 19{,}000$

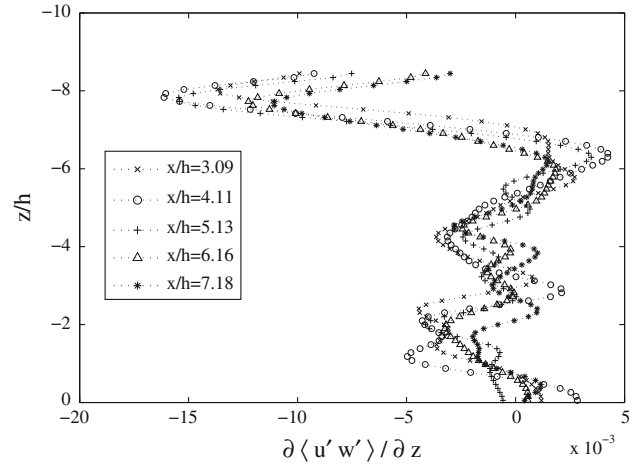

Fig. 21 Reynolds term $\frac{\partial\langle u'w'\rangle}{\partial z}$ across the *left* channel half; $y/h = 0.4; Re = 19{,}000$

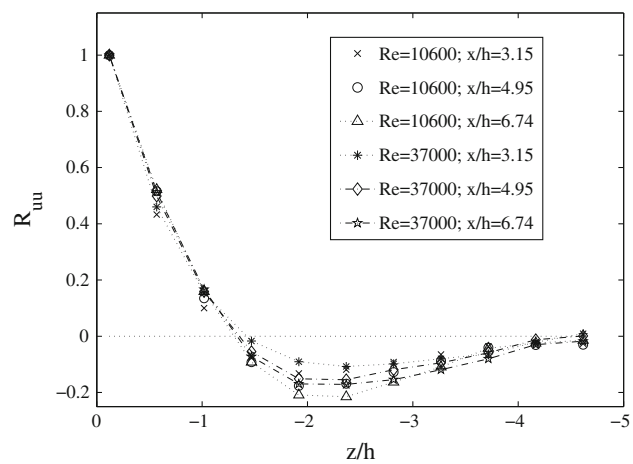

Fig. 22 Two-point correlation R_{uu} from PIV measurements; $y/h = 1.53; Re = 10{,}600$ and $Re = 37{,}000$

contribute only in a minor manner. The impact of the lateral variations on the momentum balance in the y direction was not estimated as results in the y/z plane are lacking. The terms of the momentum balance in the z direction are, of course, significantly smaller than the ones in the x.

The investigations into the homogeneity of the flow allow the following conclusions. The slight variations of the velocities across the channel width have only minor effects on the momentum balance such that the flow can be regarded as two-dimensional from $Re \geq 10{,}600$ on.

4.4 Two-point correlations

The statistical independence of a fluctuating quantity at two different positions in space can be proofed by two-point correlations. The velocity fluctuation at point 1, u_1, is independent from the respective velocity at point 2 if the two-point correlation yields zero. It is defined as:

$$R_{u_1u_2} = \frac{\langle u_1' u_2' \rangle}{\sqrt{\langle u_1' u_1' \rangle \langle u_2' u_2' \rangle}} \tag{7}$$

The distance between the $R_{u_1u_2} = 1$ and $R_{u_1u_2} = 0$, the so-called correlation length, determines half of the length of the largest flow structures. Thus, the channel is sufficiently broad if the two-point correlations diminish to zero across the half channel width. The two-point correlations were computed from the PIV data at $y/h = 1.53$ for 11 streamwise positions across about one fourth of the channel width from the center to $z/h \approx 5$. Figure 22 depicts the two-point correlations of the u component $R_{u_1u_2}$ at various streamwise positions. They become zero for all Reynolds numbers and for all streamwise positions. The correlations subside

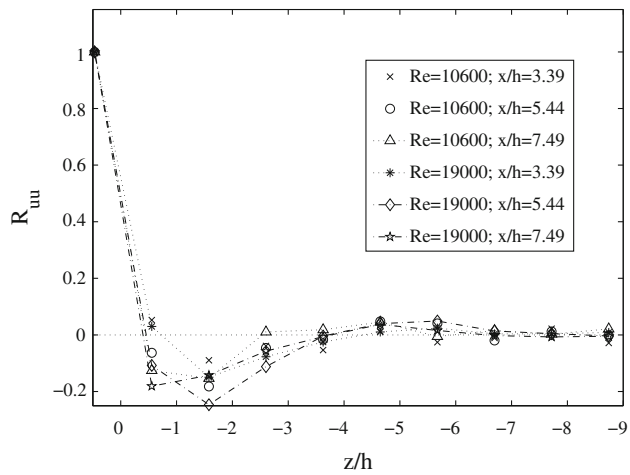

Fig. 23 Two-point correlation R_{uu} from PIV measurements; $y/h = 0.4$; $Re = 10{,}600$ and 19,000

analogously to the ones found by Fröhlich et al. (2005), who did LES at $Re = 10{,}600$ with a computational domain that was $9h$ wide. The simulated velocity fluctuations are uncorrelated after $\approx 3.5h$, whereas the experiment shows statistical independence from $4h$ on.

The distance to the correlation minimum distinguishes the mean diameter of the largest structures. It is about $2h$ and is slightly increasing with increasing Reynolds number.

From Fig. 23, it can be seen that R_{uu} does not diminish that quickly in the $y/h = 0.4$ plane in comparison with the higher level at $y/h = 1.53$. Additionally the distance of the correlation minimum decreases below $2h$. However, a secondary correlation maximum can be identified at $(y/h; z/h) = (0.4; 5.0)$. Taylor–Görtler vortices are therefore suspected to be at the windward side of the hill especially at $Re = 10{,}600$. The wavelength of these vortices of about $4.5h$ is consistent with the secondary flow structures found in the mean flow fields (compare Sect. 4.3.1).

According to the two-point correlations, the channel is sufficiently wide for the largest structures of the flow. Thus, another criteria for the homogeneity of the flow is fulfilled.

4.5 Time-averaged flow field

In the preceding sections, the measurement devices were thoroughly inspected and it was checked whether the experimental setup meets with the preconditions periodicity and homogeneity. It was proved that the flow is periodic for $Re \geq 10{,}600$ and that the lateral variations have only minor effects on the momentum balance. Therefore, the flow can be assumed to be two-dimensional to good approximation in the center of the channel. The following

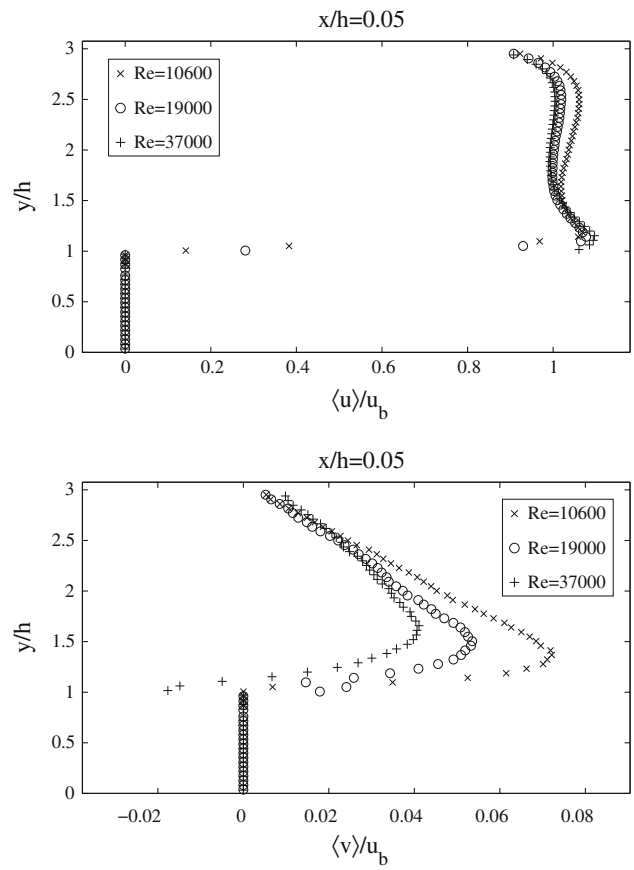

Fig. 24 Velocity profile; $x/h = 0.05$; *top* $\langle u\rangle/u_b$, *bottom* $\langle v\rangle/u_b$

assessment of first- and second-order statistics can be interpreted in terms of a flow that is periodic in the streamwise and homogeneous in the spanwise direction.

In the following, our measurements are evaluated for $Re = 10{,}600, Re = 19{,}000$ and $Re = 37{,}000$. To this end, we examine mean velocity and Reynolds stress profiles at characteristic streamwise positions.

The velocity profile of the u component develops an overshoot directly above the hill crest with increasing Reynolds number (Fig. 24). This goes in hand with a flatter profile toward the upper wall. From the velocity component in the vertical direction v, a clear tendency can be detected. The upward flow decreases with increasing Reynolds number, which indicates that the streamlines that are pointing upwards in the low Re case become flatter at the higher Reynolds number.

Next, we examine the flow profiles at $x/h = 1.0$, just after flow separation. The tilting u profile can still be seen at this position. With increasing Reynolds number, a flatter velocity profile is obtained in the upper channel third (Fig. 25). Consequently, the velocity profile is fuller above the recirculation zone at higher Re. The recirculation zone itself seems to have the same thickness and velocity profile at all Reynolds numbers. This is also the case for the

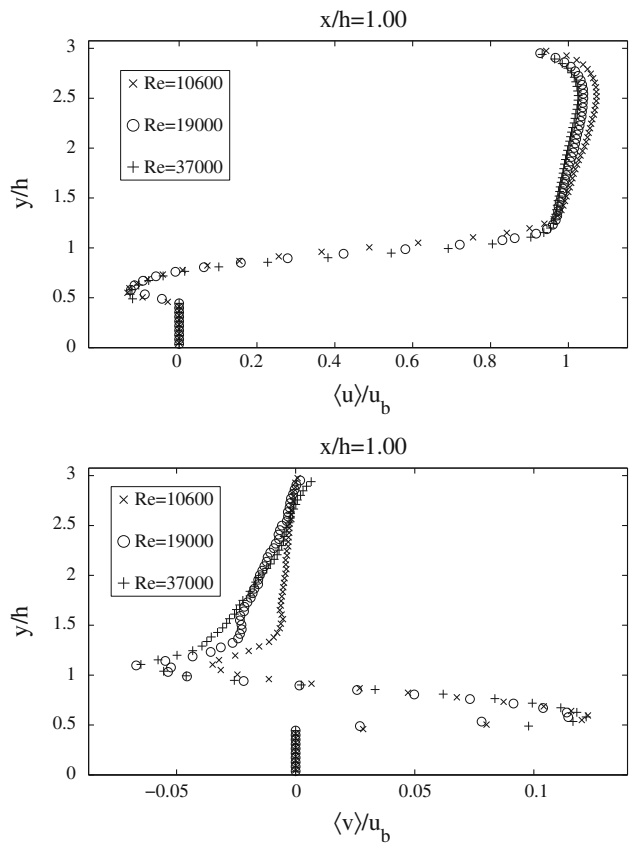

Fig. 25 Velocity profile; $x/h = 1.0$; *top* $\langle u\rangle/u_b$, *bottom* $\langle v\rangle/u_b$

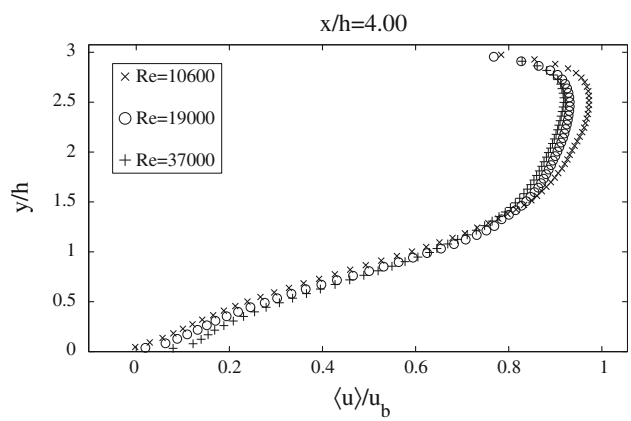

Fig. 26 Velocity profiles $\langle u\rangle/u_b$; $x/h = 4.0$

vertical velocity component, which differs between the Reynolds numbers only outside the recirculation zone in the shear layer. Directly above the recirculation zone, the flow points downwards for all Reynolds numbers, however with a vertical component increasing with Reynolds number. Thus, the momentum transfer downwards by mean convection is higher for the higher Reynolds numbers.

The position $x/h = 4$ represents the flow state close to reattachment. From the u component (see Fig. 26), it can

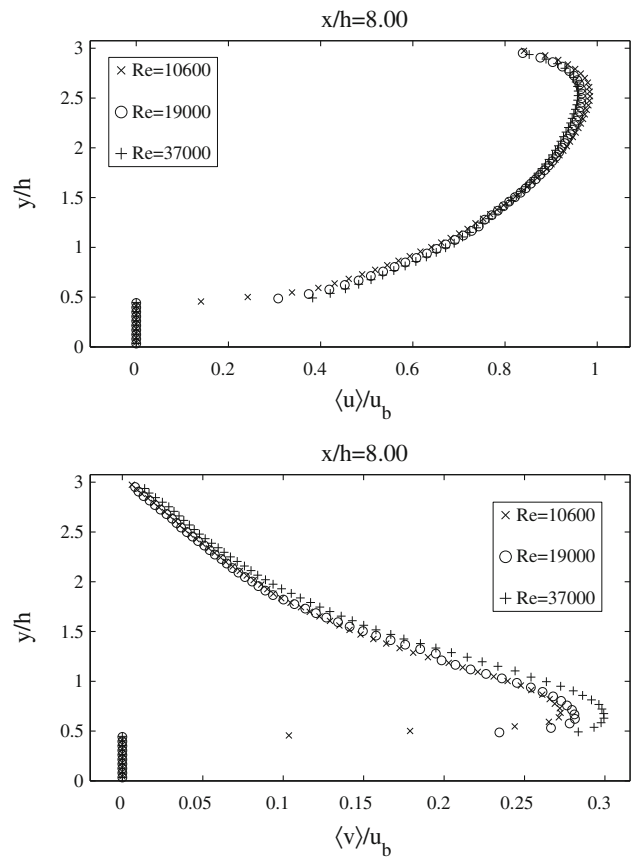

Fig. 27 Velocity profile; $x/h = 8.0$; *top* $\langle u\rangle/u_b$, *bottom* $\langle v\rangle/u_b$

be seen that the flow is detached at $Re = 10{,}600$ but has already reattached at $Re = 19{,}000$ and $Re = 37{,}000$. The attached state has a lower velocity in the upper channel half. At $Re = 37{,}000$ the flow already recovers and accelerates close to the wall. The Reynolds number effect on the mean streamwise velocity profiles is larger at this position than at the hill crest due to an increasing momentum transfer into the separation zone at the higher Reynolds numbers.

From the profiles at $x/h = 8.0$, one can guess the further development. Apparently, the Reynolds number effect on the streamwise velocity is weaker than close to reattachment as the acceleration due to the windward slope of the hill is already in effect. However, there is a considerably larger effect of the lower wall on the profile at the lower Reynolds number than at the higher ones (Fig. 27). At the lower Reynolds number, the profile goes smoothly to zero at the lower wall. This might result from higher Reynolds stresses. Because of the greater v component at higher Re, more fluid is shifted upwards (Fig. 27). This is required as due to the shorter recirculation zone and the early recovery a much higher flux is obtained at the lower channel third.

A distinct feature of the time-averaged flow field is the development of a velocity overshoot over the hill crest. Such an overshoot can be observed as well in other

experiments, such as in Motzfeld (1937), Almeida et al. (1993) or Gong et al. (1996), but not in all configurations (e.g. Hudson et al. 1996; Buckles et al. 1984). In Motzfeld's experiment, it is dependent on the shape of the hill. It appears only in the steep hill's case, while it is absent in the sinusoidal case. In Gong et al.'s experiment (Gong et al. 1996), the overshoot disappears over rough terrain. In Almeida et al.'s experiment, it appears in the single hill configuration but not in the periodic one (at least it can not be inferred from the plots presented in the paper). Almeida et al. (1993) report a speed-up factor of 2.74 for the same hill geometry as considered here. This is considerably larger than what was reported in other laboratory studies of flow over wavy walls (Athanassiadou and Castro 2001; Gong and Ibbetson 1989) or field experiments of flow over hills (Mason 1986; Zeman and Jensen 1987). In our experiment, we observe maximal velocity differences between the trough ($x/h = 6.0$) and the hill crest of $\Delta u \approx 0.8u_b$ if the vertical distance from the wall is taken as coordinate. When normalized with the local velocity at the corresponding height at $x/h = 6$, this implies to a maximum fractional speed-up ratio of about 2.8 at $Re = 10{,}600$. Almeida et al. give a comparable value of 2.74; however, they did not document on which basis this value was normalized.

Generally, the speed up of the streamwise velocity over two-dimensional hills can be predicted by asymptotic theory if the slope of the hill is not too large (Jackson and Hunt 1975; Belcher et al. 1993). In our setup, the slope of the hill is too large for the asymptotic theory to be applicable (Belcher et al. 1993). Some ingredients may however be useful for interpreting our results, especially the strong velocity overshoot and its development with Reynolds number. In our case, the streamlines are strongly compressed at the windward side of the hill. Therefore, we estimated the speed up S (Eq. 8) in our configuration along streamlines, i.e. at constant values of stream function Ψ between the positions $x/h = 6.0$ and $x/h = 0.05$, of which the first is the position at which the streamlines reach closest down to the wall and the second is above the hill crest.

$$S(\Psi) = \frac{\langle u\rangle(x/h = 0.05, \Psi) - \langle u\rangle(x/h = 6.0, \Psi)}{u_b} \qquad (8)$$

The result is displayed in Fig. 28. When approaching from top toward the bottom wall, we run through a region $1 \leq \Psi \leq 1.75$ in which the speed up is nearly constant. This can be interpreted in terms of an overall acceleration of the flow due to the blockage of the channel. Below that layer, the speed up increases and reaches a maximum before it decreases toward the no-slip wall. According to asymptotic analysis, the maximum separates the inner layer from the outer layer. In the inner layer, the stress gradients are important for the flow perturbations, and in the outer layer, they are largely affected by inviscid dynamics, even if the unperturbed flow is determined by viscous or Reynolds stresses (Belcher et al. 1993). In our case, the speed up is remarkably unaffected by Reynolds number in the outer layer. This confirms the assumption that the perturbation in this layer is mainly governed by inviscid dynamics. In the inner layer, the shear stress dynamics affect the perturbation of the velocity profiles. Its thickness is of order u_τ/u_b. Therefore, this layer is strongly Reynolds number dependent, as can be seen in our results. The inner layer thickness is considerably thinner at the highest Reynolds number than in the lower ones, which explains why the velocity overshoot over the crest of the hill increases with the Reynolds number.

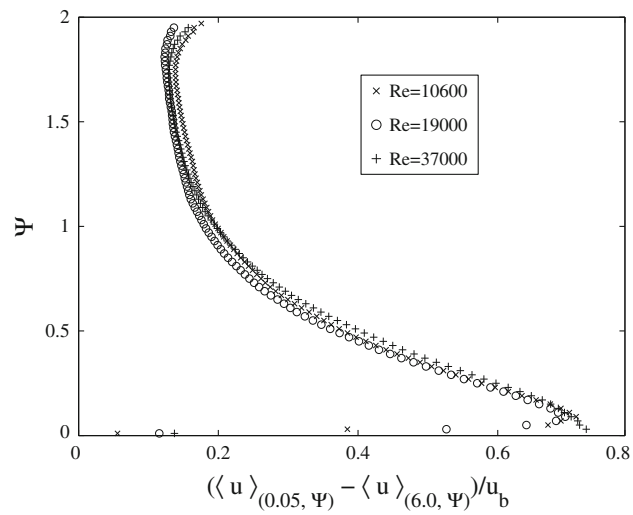

Fig. 28 Speed-up profile at the hill crest with respect to $x/h = 6.0$, normalized by u_b. The *vertical axis* depicts the stream function

4.6 Reynolds stresses

The PIV Reynolds stresses $\langle u'u'\rangle/u_b^2$ close to the wall at $x/h = 0.05$ were substituted by LDA measurements because only a poor quantity of reliable vectors was obtained there; these numbers offer sufficient statistics for the first-order but not for the second-order moments. Another argument for using the LDA measurements at these particular points is that the vertical extent of the LDA measurement volume is a tenth smaller than the PIV interrogation areas. Therefore, the fluctuations can be resolved more accurately.

The streamwise Reynolds stresses develop two maxima over the crest ($x/h = 0.05$; Fig. 29). The secondary maximum at $y/h \approx 1.7$ appears in all measured stress components. Fröhlich et al. (2005) conclude that these intense stresses are being convected from upstream positions. The vertical stress level partly exceeds the streamwise one. All stresses decrease with increasing Reynolds number in the

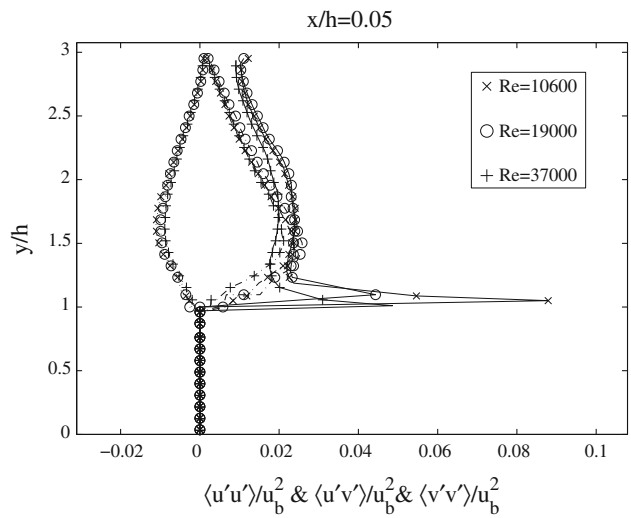

Fig. 29 Reynolds stress profiles; $x/h = 0.05$; $\langle u'u'\rangle/u_b^2$, $\langle u'v'\rangle/u_b^2$ and $\langle v'v'\rangle/u_b^2$

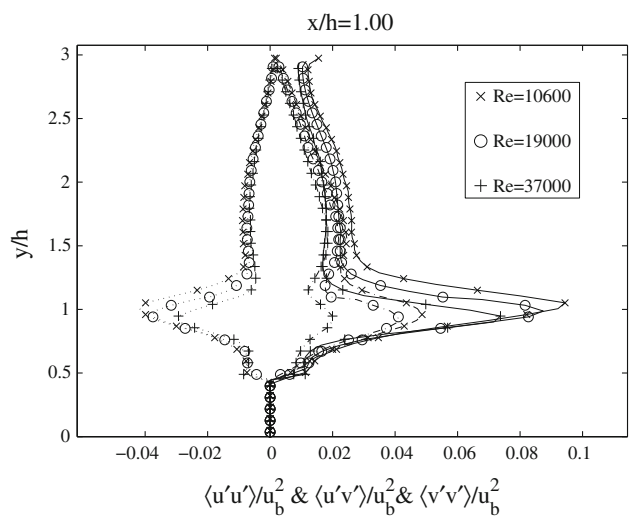

Fig. 30 Reynolds stress profiles; $x/h = 1.0$; $\langle u'u'\rangle/u_b^2$, $\langle u'v'\rangle/u_b^2$ and $\langle v'v'\rangle/u_b^2$

secondary maximum which is in the bulk of the flow. The sharp peak directly at the wall ($y/h < 1.1$) appearing in $\langle u'u'\rangle$ does not have a counterpart in the vertical or shear stresses. Note that for the lower two Reynolds numbers, this near-wall peak is spatially resolved while it is not for $Re = 37{,}000$. Therefore, the rms in this near-wall peak seem to be lowest at the highest Reynolds number which would be in contradiction with the fact that the velocity gradient is steepest. However, the value keeps on increasing when approaching the wall. This near-wall peak is found in (Fröhlich et al. 2005) or (Breuer et al. 2009) as well, but the LESs yield even higher peaks. Obviously, the flow is strongly anisotropic and not in equilibrium. After

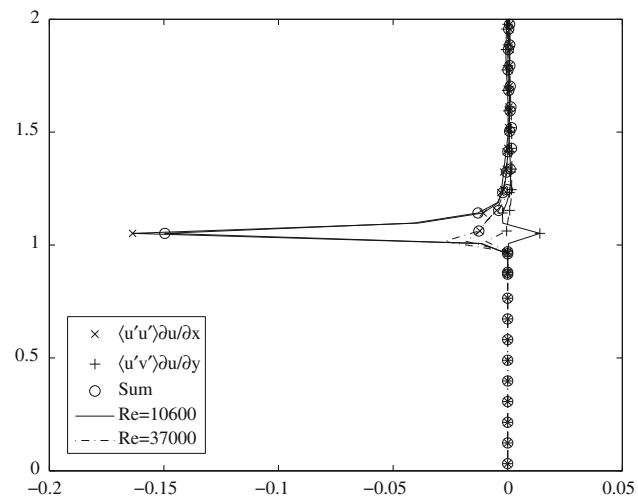

Fig. 31 Production of streamwise Reynolds stress; $x/h = 0.05$; $\langle u'u'\rangle\frac{\partial\langle u\rangle}{\partial x}$, $\langle u'v'\rangle\frac{\partial\langle u\rangle}{\partial y}$ and sum normalized by u_b^3/h

separation, the steep near-wall peak develops into one located in the separated shear layer (Fig. 30).

Downstream of separation at $x/h = 1.0$, the shear layer gives rise to instabilities that grow fast and produce considerable stress levels in all components at $y/h \approx 1.0$ (Fig. 30). The peak values are lower at the higher Re like before separation, an effect that is strongest in $\langle v'v'\rangle/u_b^2$. At this streamwise position, the secondary maximum can still be observed which can be explained by convection of the turbulence from upstream at $y/h \approx 1.5$.

The production of turbulent kinetic energy in the wall-normal component is generally weak, as already observed by Fröhlich et al. (2005). They interpret the strong peak in streamwise stresses at the crest ($x/h = 0.05$) as being generated by the high shear strain in the boundary layer. Our measurements however let us conclude that turbulence is produced in the streamwise component by the term $\langle u'u'\rangle\frac{\partial\langle u\rangle}{\partial x}$ as a result of the strong deceleration of the near-wall velocity overshoot before flow separation (Fig. 31). At $Re = 10{,}600$, this happens more upstream than at $Re = 37{,}000$ at which similar high values are obtained at $x/h = 0.5$ (not shown). In the further development in the separated shear layer ($x/h = 1.0$, Fig. 32), the dominant production term is $\langle u'v'\rangle\frac{\partial\langle u\rangle}{\partial y}$ which can be attributed to Kelvin-Helmholtz vortices developing in the shear layer as observed by Fröhlich et al. (2005). Only a minor part of the turbulent kinetic energy is produced in the vertical component which, together with the spanwise component, obtains the major fraction of its energy by pressure strain correlation. There is a notable difference in the main production term between $Re = 10{,}600$ and $Re = 37{,}000$. At the higher Reynolds number, the shear layer is considerably thinner leading to a thinner peak of the production

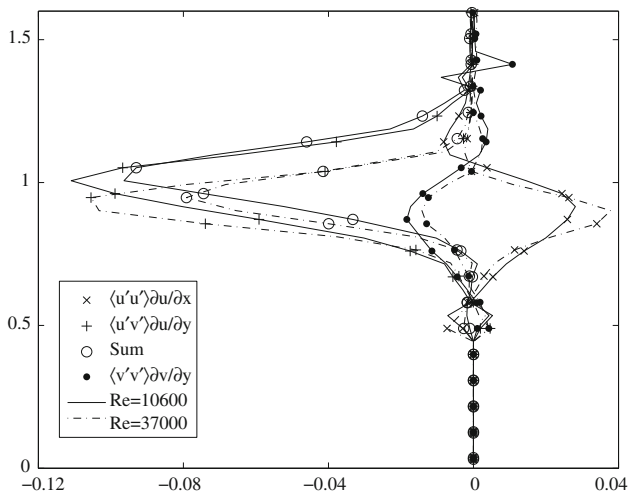

Fig. 32 Production of streamwise and vertical Reynolds stress; $x/h = 1.0$; $\langle u'u'\rangle\frac{\partial\langle u\rangle}{\partial x}$, $\langle u'v'\rangle\frac{\partial\langle u\rangle}{\partial y}$, sum and $\langle v'v'\rangle\frac{\partial\langle v\rangle}{\partial y}$ normalized by u_b^3/h

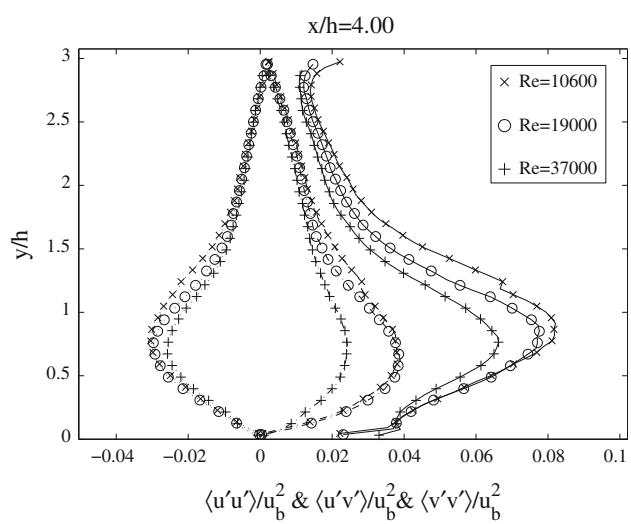

Fig. 33 Reynolds stress profiles; $x/h = 4.0$; $\langle u'u'\rangle/u_b^2$, $\langle u'v'\rangle/u_b^2$ and $\langle v'v'\rangle/u_b^2$

term. Together with a maximum that is by 20% lower than at the lower Reynolds number, the total production of turbulent kinetic energy at this position is lower at the higher Reynolds number leading to lower Reynolds stresses.

The rms of the streamwise velocity component shows the same behavior close to reattachment ($x/h = 4.0$, Fig. 33) as upstream. Particularly in $Re = 37{,}000$, the components of $\langle u'u'\rangle/u_b^2$ and $\langle v'v'\rangle/u_b^2$ are notedly smaller than for $Re = 10{,}600$ and $Re = 19{,}000$. The vertical fluctuations $\langle v'v'\rangle/u_b^2$ at $y/h \approx 0.75$ lose half of their magnitude between $Re = 10,600$ and $Re = 37{,}000$.

The lower Reynolds stresses at higher Reynolds numbers persist even at the upwind side of the hill where the flow is accelerated and pointing upwards ($x/h = 8.0$,

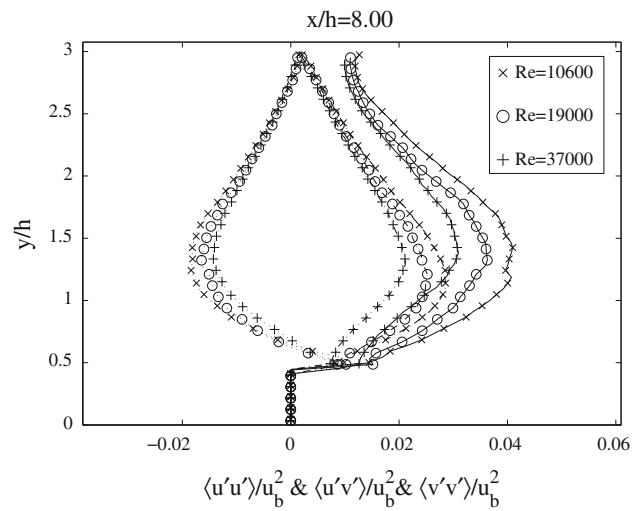

Fig. 34 Reynolds stress profiles; $x/h = 8.0$; $\langle u'u'\rangle/u_b^2$, $\langle u'v'\rangle/u_b^2$ and $\langle v'v'\rangle/u_b^2$

Fig. 34). A broad peak is found there for all *Re* which corresponds to the thickness of the shear layer at that position. This maximum will form the secondary maximum further downstream (between $x/h = 0.05$ and $x/h = 1.0$). A similar behavior was already observed by Buckles et al. (1984). The Reynolds number dependence of the Reynolds stresses contribute to the observed changes in the mean velocity profiles with Reynolds number. The lower Reynolds shear and normal stresses at the higher Reynolds number result in a reduced momentum transfer across the streamlines which makes the profiles fuller at the lower wall. These fuller profiles represent a higher kinetic energy of the near-wall streamlines. Together with a thinner inner layer, in which the development of the Reynolds stresses is important for the local flow acceleration, it gives rise to larger near-wall peaks in the further development of the flow above the hill crest (compare $x/h = 0.05$, Fig. 24). It is interesting to note that most of the LES models fail in reproducing the Reynolds number dependence of the Reynolds stresses and prediction of the velocity peak above the hill crest (Manhart et al. 2009).

4.7 Separation zone

The prediction of the reattachment point of a separated flow is hitherto a challenging task for numerical methods. This is one of the reasons why the present case has been selected by several research collaborations as a benchmark for numerical models. In a turbulent flow, the reattachment line fluctuates strongly in space and time. Large-scale and low-frequency oscillations have been observed in many experiments and numerical simulations (Adams and Johnston 1988; Na and Moin 1998; Patrick 1987). Similar effects are present in our configuration. The recirculation

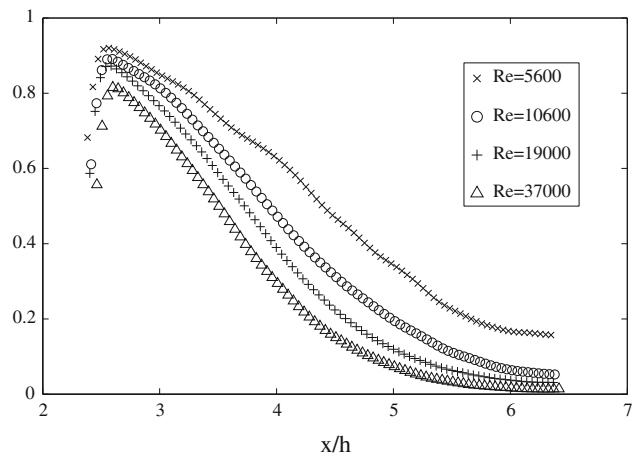

Fig. 35 Fraction of time during which velocity vector nearest to the wall points upstream (reverse flow factor)

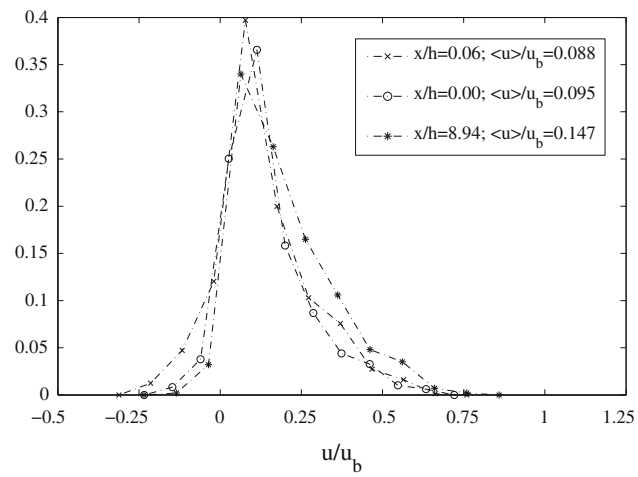

Fig. 36 Comparison of u/u_b from LDA measurements by means of PDF; $y/h = 1.01; Re = 10{,}600$

Table 5 Mean point of reattachment in dependence of the Reynolds number

Re	*x/h*
5,600	4.83
10,600	4.21
19,000	3.94
37,000	3.76

bubble breaks up frequently and structures drift along the channel bottom. In most of the instantaneous flow fields, we observed that the velocity nearest to the wall changes its sign three times along the bottom wall. We estimated the fraction of time during which this velocity is pointing upstream in Fig. 35. At $Re = 5{,}600$ and 10,600, instantaneous backflow does not disappear after reattachment, showing nearly 20% backflow at the lowest Reynolds number. In turn, at the highest Reynolds number, the flow is forward for at least 20% of time throughout the whole bottom wall, even in the mean separation region. This indicates large-scale structures crossing the whole backflow region reaching down to the wall as observed e.g. by Manhart and Friedrich (2002).

The time mean point of reattachment determined by PIV measurements is listed for the four Reynolds numbers investigated in Table 5. These points were found as follows. From the mean values of the u component, the two neighboring interrogation areas were selected, where the flow direction changes from negative to positive values. From a linear interpolation between the two velocities, the point with the velocity zero was obtained. This was done for the interrogation area row closest to the wall and the layer above. The extrapolation of the zero velocity line yielded the point of reattachment. It has to be noted that this procedure estimates the point of zero wall shear stress by interpolation in the streamwise and the vertical direction. It yields later reattachment locations as the zero velocities in the center of the wall nearest interrogation area would give. Therefore, the locations of 50% reverse flow in Fig. 35 are a little bit more upstream than the points of zero wall shear stress.

The mean point of reattachment moves upstream with growing *Re* but eventually a Reynolds number–independent state will be reached at some higher *Re* than the ones considered. This is in line with observations in other flow configurations such as backward facing step by Armaly et al. (1983). The fact that the flow reattaches earlier at higher *Re* cannot be explained by a higher momentum transfer toward the separation zone by the Reynolds stresses as they tend to be lower at higher *Re*. We observe, however, that close to the separation point at $x/h = 0.05$ (Fig. 24), the streamlines point downwards immediately above the crest at $Re = 37{,}000$ while they point upwards at the lower Reynolds numbers. We also observe that the momentum and the shear rate above the separation region directly after separation increase with Reynolds number. This indicates a considerable influence of the velocity speed up above the crest on the reattachment length.

4.8 Recirculation on hilltop

Breuer et al. (2009) found a tiny recirculation zone on the hill crest in time average in an LES at $Re = 10{,}595$. To identify it in the experiment, LDA measurements were undertaken because they reach closer to the wall (<1 mm) than the PIV measurements. Figs. 36 and 37 show the PDFs of the u component at the wall closest measuring point at $y/h \approx 1.01$ and at different x/h positions. The average is positive at every position and Reynolds number.

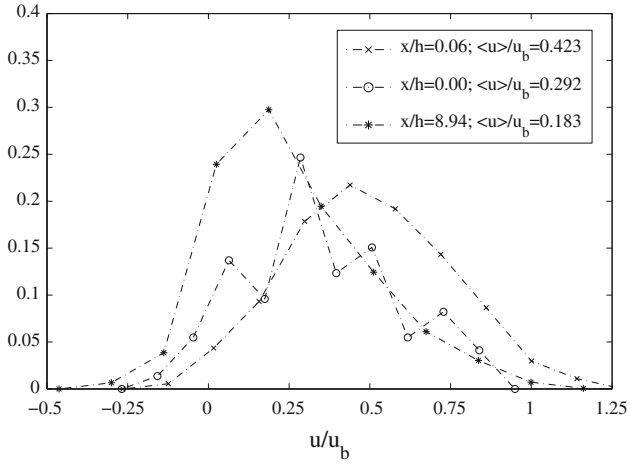

Fig. 37 Comparison of u/u_b from LDA measurements by means of PDF; $y/h = 1.01; Re = 37{,}000$

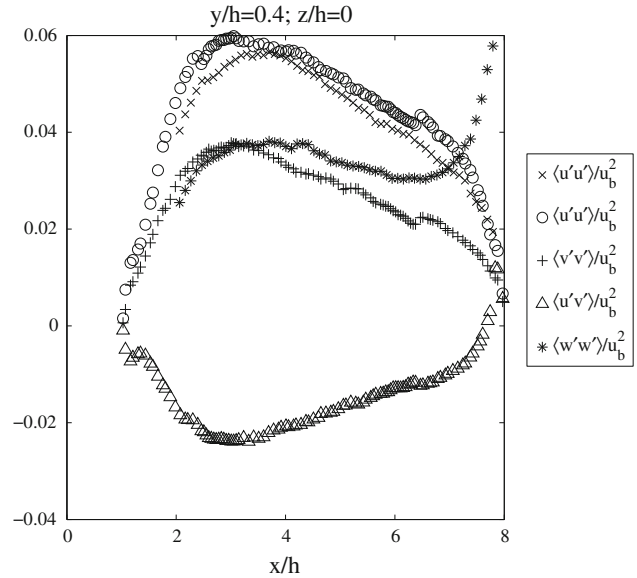

Fig. 38 Comparison of $\langle u'u'\rangle/u_b^2$ (from *vertical* and *horizontal* laser sheets), $\langle v'v'\rangle/u_b^2$ and $\langle u'v'\rangle/u_b^2$ with $\langle w'w'\rangle/u_b^2$, $y/h = 0.40$, $z/h = 0; Re = 10{,}600$

However, depending on the position, for a not negligible fraction of time reverse flow occurs on the hill crest. If the PDF is regarded as the velocity distribution across the measurement volume in the y direction the tiny recirculation zone on top of the hill crest will be probable. Additionally, one has to account for the fact that the Gaussian intensity distribution of the laser beam leads to fewer interference maxima at the fringe of the measurement volume resulting in fewer valid bursts there. Nevertheless, a wall closer measurement was not possible.

According to the experiments undertaken, the recirculation zone on the hill crest could at least not be excluded. In any case, incipient detachment was found. Furthermore, according to the experiments, the incipient detachment travels upstream for higher Reynolds numbers.

The PDFs at $Re = 37{,}000$ appear broader than the respective functions for $Re = 10{,}600$ which comes along with a higher rms value ($\mathrm{rms}_{37{,}000} \approx 0.06$ and $\mathrm{rms}_{10{,}600} \approx 0.02$). This fact seems to be in contradiction to the general observation of decreasing rms values with increasing Re. However, the peak of the rms value at $Re = 37{,}000$ (Fig. 29) is steeper and closer to the wall and therefore not resolved by these rms profiles. For lower Re, this peak in $\langle u'u'\rangle/u_b^2$ has already decreased at the wall nearest measurement point.

4.9 Splatting effect on the windward side of the hill

Fröhlich et al. (2005) discovered an exceptionally high fluctuation level in the spanwise velocity component on the windward side of the hill. They explain this issue with the splatting effect, i.e. turbulence balls hit the obstacle and break apart to either side. Similarly, high spanwise fluctuations were also observed in other reattaching flows e.g. (Castro and Haque 1987; Manhart 1998).

In the horizontal laser sheets, we have access to the spanwise fluctuations. We plot all non-negative components of the Reynolds stress tensor along the intersection line between the vertical and horizontal laser sheets. Thus, we are able to cross-check our results for the streamwise Reynolds stress. We also observe the splatting effect in the experiment for $Re = 10{,}600$ (see Fig. 38). When approaching the hill surface from upstream in the x direction, the spanwise fluctuations $\langle w'w'\rangle$ increase dramatically shortly before the hill is reached. They reach their maximum value, which is much higher than the fluctuation levels of the other velocity components, right in front of the hill. This can typically be attributed to the splatting effect.

The results for $Re = 19{,}000$ at the same position, however, show a different behavior (Fig. 39). The spanwise fluctuation level starts increasing at $x/h = 6$ like in the lower Reynolds number. It even attains higher values than the streamwise one at $x/h = 7$ but then is damped toward the wall like the other stress components. We have checked this result by various methods. The number of outliers is not higher than 3% in this area, and the statistics do not vary between interpolated vectors included or excluded. All components of the Reynolds stresses comply qualitatively between both Reynolds numbers plotted except the spanwise normal stress. In addition, we made use of the continuity equation to estimate the magnitude of $(\partial w'/\partial z)^2$ in the vertical laser sheets. This quantity rises toward the wall in the same manner in both Reynolds numbers. We conclude therefore that the damping of the spanwise fluctuations toward $x/h = 8$ in Fig. 39 has to be taken with care.

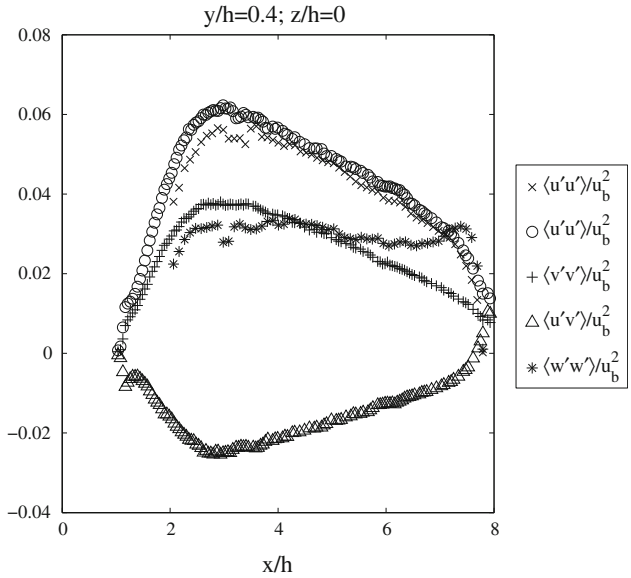

Fig. 39 Comparison of $\langle u'u'\rangle/u_b^2, \langle v'v'\rangle/u_b^2$ and $\langle u'v'\rangle/u_b^2$ with $\langle w'w'\rangle/u_b^2, y/h = 0.40, z/h = 0; Re = 19,000$

5 Conclusions

An experiment has been thoroughly set up in the Hydromechanik Laboratorium of the Technische Universität München. The flow over periodically arranged hills has been investigated for Reynolds numbers ranging from 5,600 up to 37,000. PIV field data were validated by pointwise LDA measurements. Ten hills were implemented in order to reach periodic flow conditions in the streamwise direction. This periodicity in the streamwise direction was shown for $Re \geq 10{,}600$ for hills six to eight. The flow variations in the spanwise direction decrease with increasing Reynolds number, and they were identified to have minor effects on the momentum balance. The influence of the side walls extends to approximately four hill heights into the domain, thus leaving an area of about ten hill heights unaffected. The flow can therefore be considered as nominally two-dimensional in the center of the channel for $Re \geq 10{,}600$.

Flow characteristics were investigated, and the development of the flow with increasing Reynolds number was discussed. A tiny incipient detachment was found on the hill crest which moves upstream with increasing Re. The flapping effect was found on the windward side of the hill for $Re = 10{,}600$ but not as strong for $Re = 19{,}000$.

The large recirculation zone behind the hill flattens and the mean point of reattachment moves upstream with increasing Reynolds number. We found that for all Reynolds numbers, there is no location in the mean separation bubble in which the instantaneous flow is always negative. Only for the highest Reynolds number, instantaneous flow reversal disappears after mean reattachment on the ground plate which could be a consequence of the shorter recirculation zone. The flatter recirculation zone at the higher Reynolds number is connected to increasing momentum transfer by mean convection rather than by Reynolds stresses.

It is worth noting that the Reynolds stresses in the bulk of the flow have been found to decrease with increasing Reynolds number. This is of special impact on the momentum transfer across the streamlines close to the windward side of the hill. In this area, the flow is strongly accelerated by the constriction of the channel by the hill. As the Reynolds stresses—normal and shear—are larger at the lower Reynolds numbers, we find a smoother velocity profile close to the wall at the lower Reynolds number and a loss of momentum of the near-wall streamlines. This comes together with a thinner inner layer in which the development of the Reynolds stresses is of importance for the local acceleration of the flow. Outside of this layer, the flow speed up has been found to be independent of Reynolds number which is in line with the assumption of inviscid mean flow acceleration usually made in asymptotic theories for flows over small hills. Due to the thinner inner layer, the speed up over the hill reaches higher values in higher Reynolds numbers. This explains the increasing velocity overshoot over the crest of the hill with increasing Reynolds number. It is obvious that the high near-wall peak at $Re = 37{,}000$ can only be predicted by a numerical method if the correct Reynolds stresses are predicted at the windward side of the hill.

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